

AI, ML & Vibe Coding

Introductory course for IT professionals



DGS presenta: AI, Machine Learning e Vibe Coding Introduzione



di Antonio Piemontese
Settembre 2026

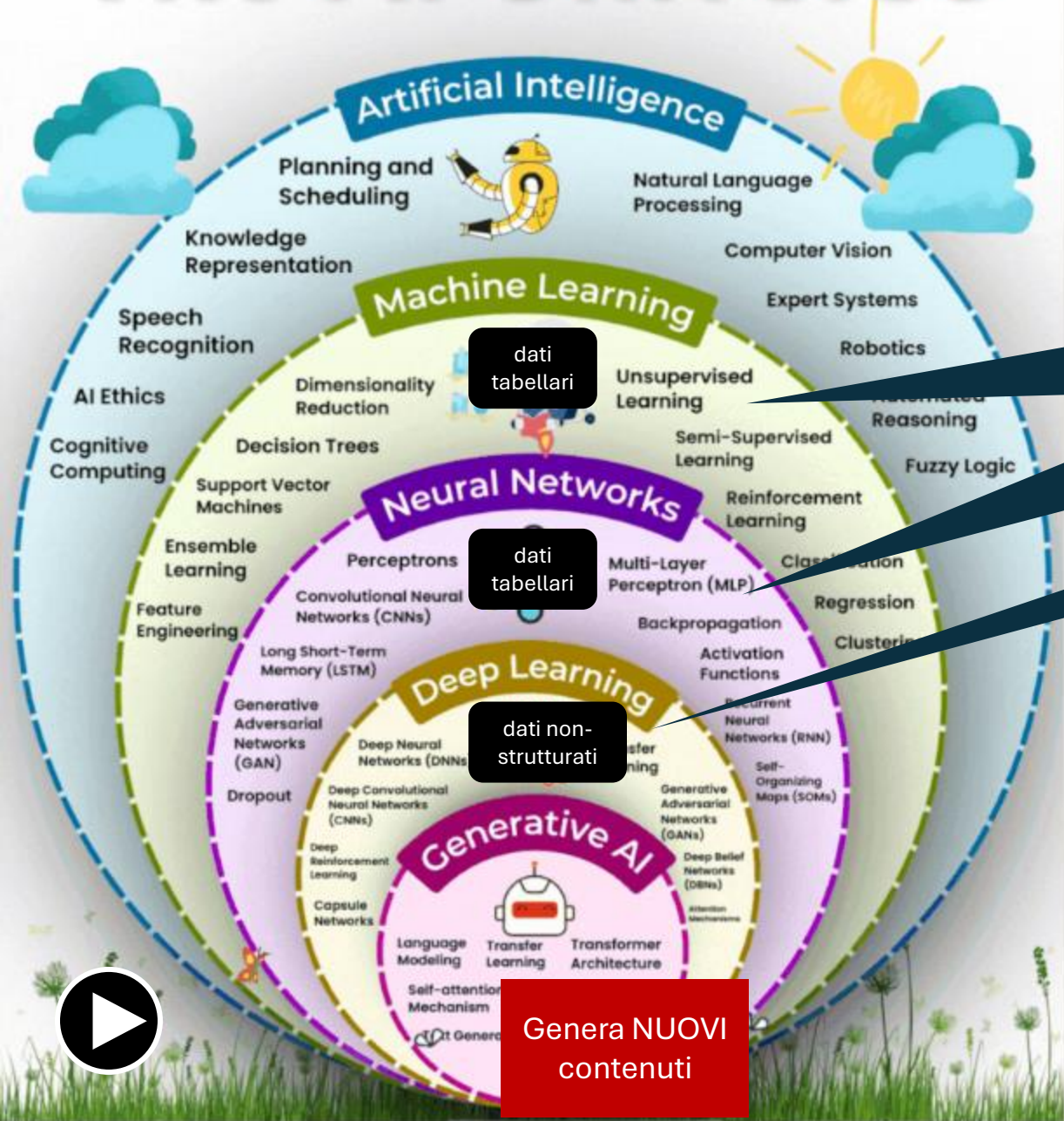


A word cloud centered around the terms 'ML AI' and 'Gen AI'. The words are arranged in a circular pattern, with 'ML AI' and 'Gen AI' being the largest and most prominent. Other words include 'Deep Learning', 'LLM', 'Chatbot', 'NLP', 'Neural Networks', 'Transformers', 'CNN', 'RNN', 'GAN', 'Perceptron', 'MLP', 'VS Code', 'Cursor', 'IDE', 'Theory', 'Practice', 'GitHub Copilot', 'Time Series', 'Prompt Engineering', 'Unsupervised Learning', 'Reinforcement Learning', 'Supervised Learning', 'Text Mining', 'Clustering', 'Classification', 'Regression', 'MLOps', and 'Perception'. The colors range from light blue to dark purple.

CNN RNN
Transformers Neural Networks GAN
MLOps Deep Learning Perceptron
VS Code ML AI LLM MLP
Cursor IDE Theory
Regression Practice
Classification GitHub Copilot
Clustering NLP Chatbot Time Series
Text Mining Prompt Engineering
Supervised Learning Unsupervised Learning
Reinforcement Learning

Introduzione alla AI generativa

The AI Universe



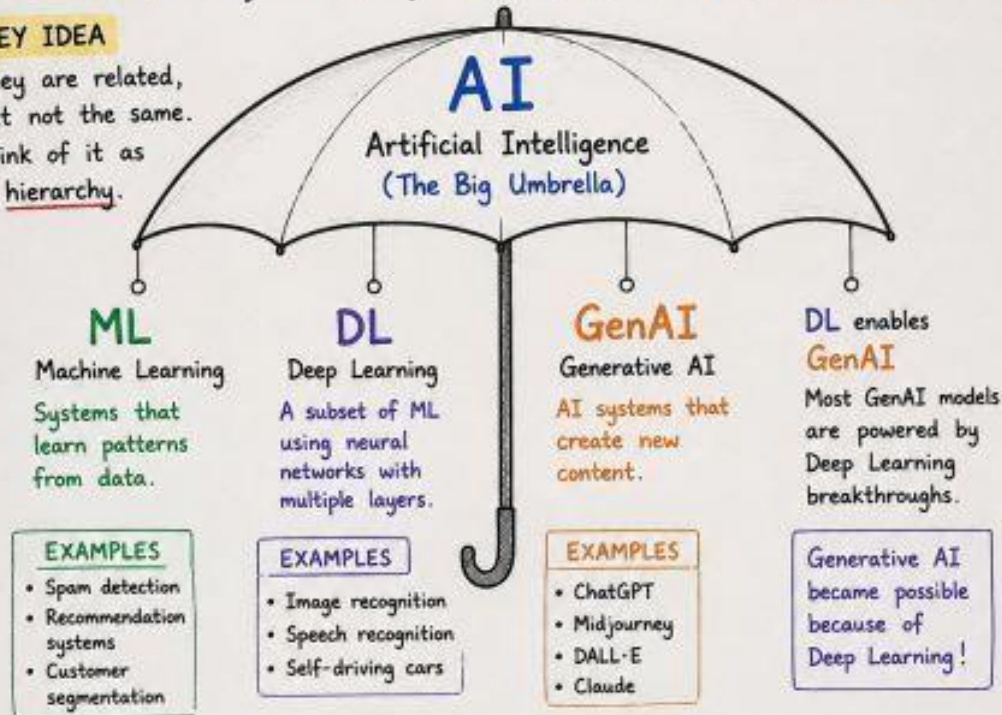
Capire i dati aziendali,
Prevedere i fenomeni di
business («analisi predittive»),
Decisioni operative

Generare
contenuto
NUOVO

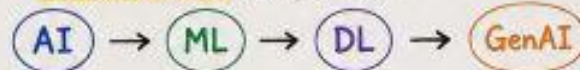
DIFFERENCE BETWEEN AI, ML, DL & GenAI

KEY IDEA

They are related,
but not the same.
Think of it as
a hierarchy.



EVOLUTION (Simple Flow)



REAL-WORLD EXAMPLES

- AI → Recommendation systems
- ML → Spam detection
- DL → Image recognition
- GenAI → ChatGPT, Midjourney, Claude

WHY THIS MATTERS

- ✓ Learn AI faster
- ✓ Choose the right tools
- ✓ Build better projects
- ✓ Understand modern AI systems



The future of work
will involve AI collaboration
everywhere.



SAVE THIS POST
if you're learning AI!

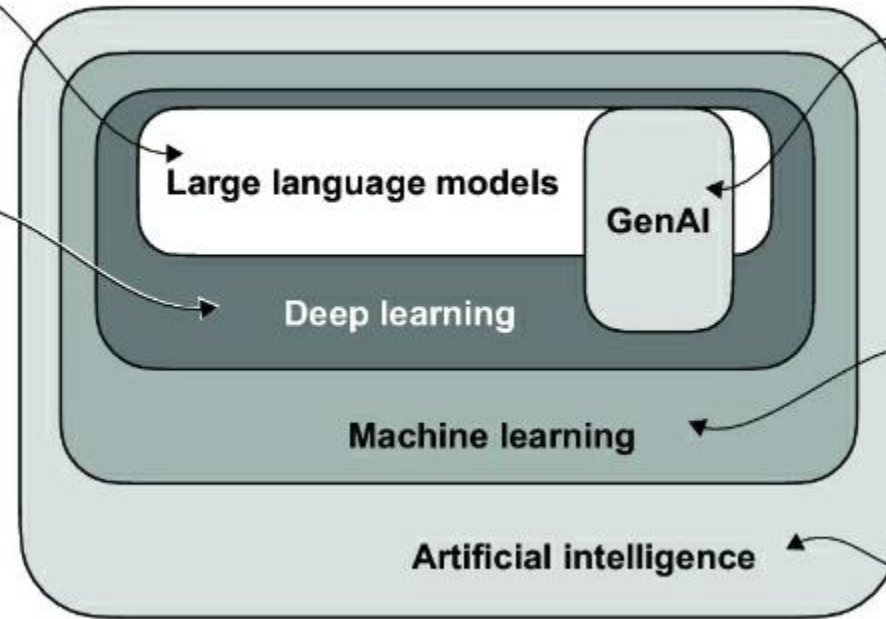


WHICH AI TOPIC
should I simplify next?



Deep neural network for parsing and generating human-like text

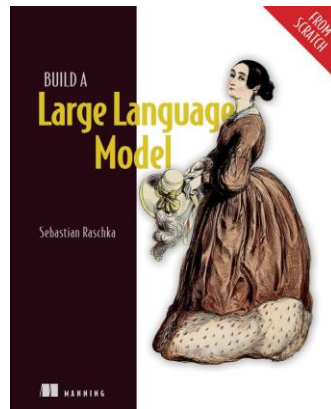
Machine learning with neural networks consisting of many layers



GenAI involves the use of deep neural networks to create new content, such as text, images, or various forms of media

Algorithms that learn rules automatically from data

Systems with human-like intelligence



Tilly Norwood, l'attrice che non esiste perché creata con l'intelligenza artificiale. Ma è già stata ingaggiata da un'agenzia di talenti

[link](#)

di Walter Veltroni

Fa discutere la prima star realizzata con l'AI. Dubbi sul futuro di Hollywood



Eva Murati, la prima attrice italiana realizzata con l'intelligenza artificiale debutta a Roma

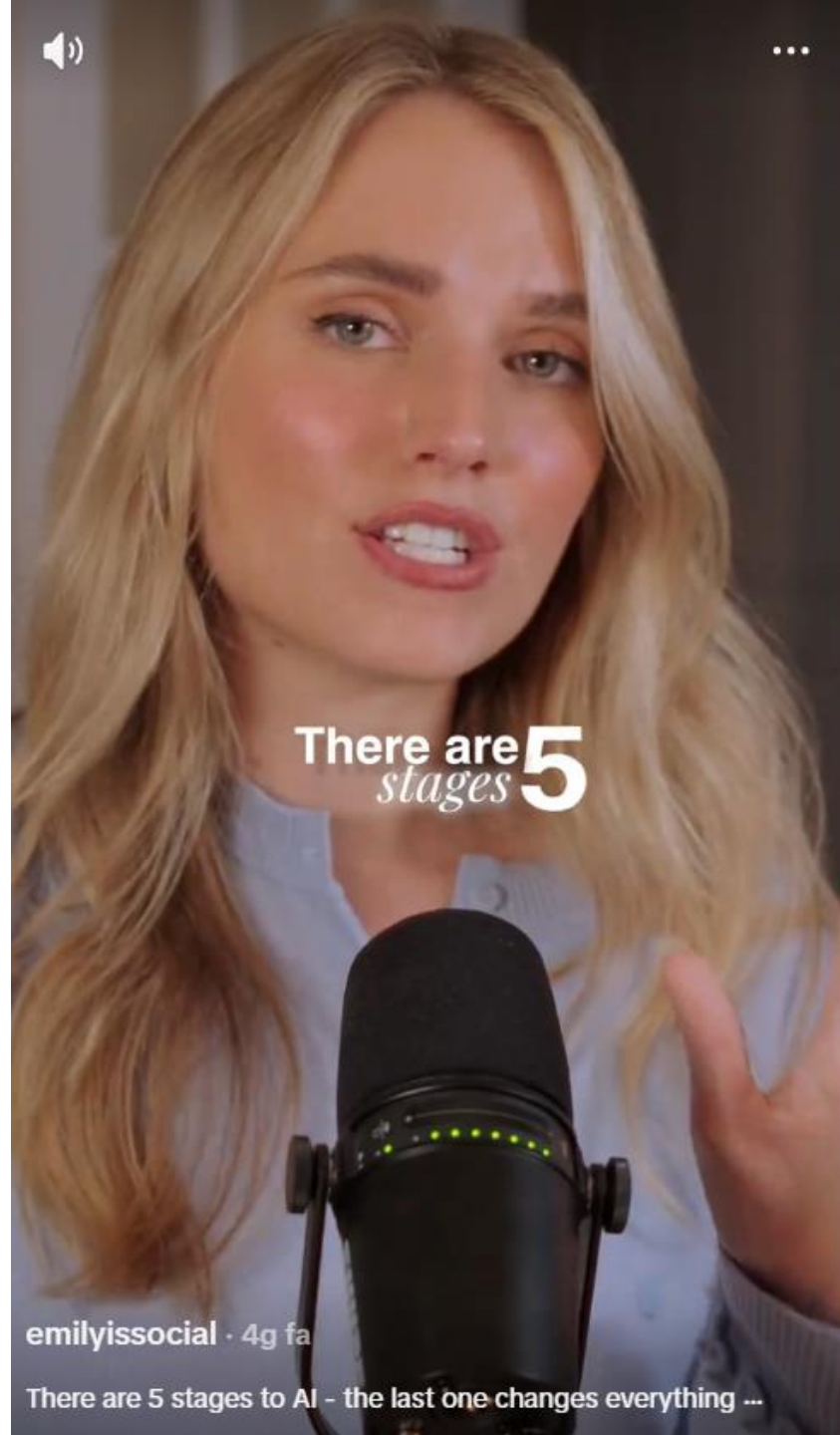
di Cristina Marrone

[link](#)

La finta attrice è protagonista di The Last Image, cortometraggio italiano realizzato con l'AI generativa grazie al lavoro di 50 persone



Non è una buona
presentazione della AI
generativa, anche se molto
diffusa



play del
video

emilyissocial · 4g fa

There are 5 stages to AI - the last one changes everything ...

Mit: 95% of GenAI business projects do not produce value, only 5% bring returns

Unofficial AI use among employees exceeds licences, influencing job change more than official projects. This is the finding of the report 'State of AI in Business 2025', compiled by the research team Mit Nanda (Networked Agents and Decentralised AI) on more than 300 AI initiatives, 52 company interviews and 153 questionnaires collected at industry conferences

20 August 2025

Despite investments of between USD 30 and 40 billion by companies, 95 per cent of **generative artificial intelligence (GenAI)** projects do not produce any concrete impact on company accounts. This is the finding of the report '**State of AI in Business 2025**', compiled by the **Mit Nanda** (Networked Agents and Decentralised AI) research team on more than 300 AI initiatives, 52 company interviews and 153 questionnaires collected at industry conferences.

The research highlights a clear divide, dubbed the 'GenAI Divide': only five per cent of integrated pilot projects generate value, with returns in the millions, while the vast majority remain stuck at the starting line, with no effect on company accounts. It is not 'model quality' or 'regulation' that are the determining factors, according to the research, but the approach taken by companies.

Reduced testing

Popular tools such as ChatGPT and Copilot have exceeded 80% adoption and nearly 40% deployment, but they only affect individual productivity, not operating margin. Customised enterprise systems, on the other hand, are often rejected: 60% of companies evaluate them, **but only 20% get to test them** and barely 5% take them into production. The main reasons are the difficulty of integrating them into workflows, the lack of contextual learning and the lack of adaptability.

The phenomenon is reflected at sector level: out of nine sectors analysed, only technology and media show signs of structural transformation. **In healthcare, finance, retail and manufacturing, adoption remains superficial, with experiments without any particular effect.** Concrete savings emerge mainly from the reduction of external service costs: between USD 2 and 10 million per year in customer service and document processing and up to 30 per cent less costs for creative agencies.

An executive of a medium-sized manufacturing company summed it up like this: 'The enthusiasm on LinkedIn says that everything has changed, but in our operations nothing is really different. We process some contracts faster, but that's about it'.

STUDIO MIT NANDA

AI, in azienda è un flop? Ma il successo è possibile, ecco come

Home > Industry 4.0/Innovazione In Azienda



Secondo il report The GenAI Divide: State of AI in Business 2025, realizzato dal MIT (Project NANDA) solo il 5 per cento dei progetti di intelligenza artificiale porta vantaggi alle aziende. Ma a ben leggere, è una questione non tecnologica, ma di buone scelte da fare e di cultura

Pubblicato il 25 ago 2025

Maurizio Carmignani

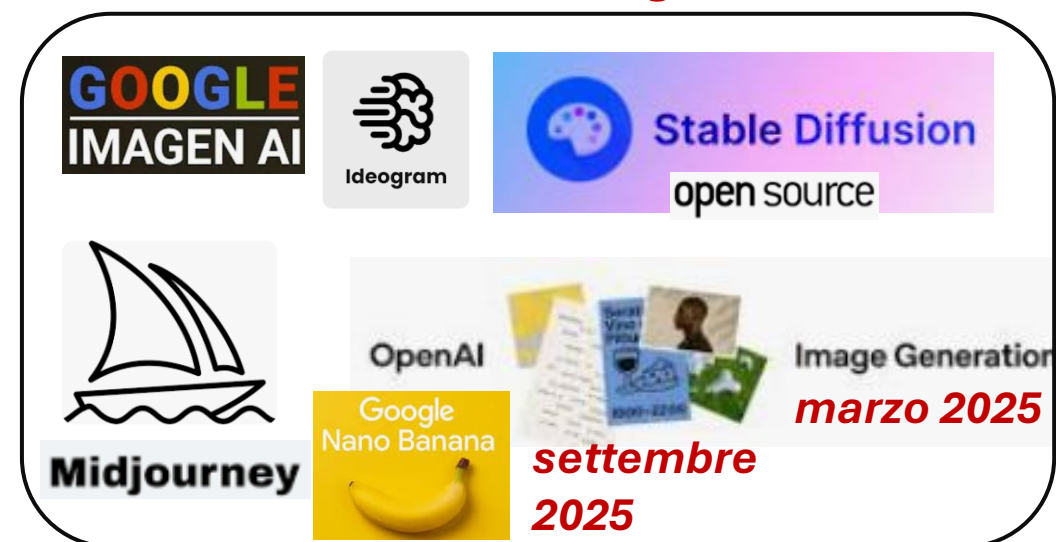
Founder & CEO – Management Consultant, Trainer & Startup Advisor

link
all'articolo

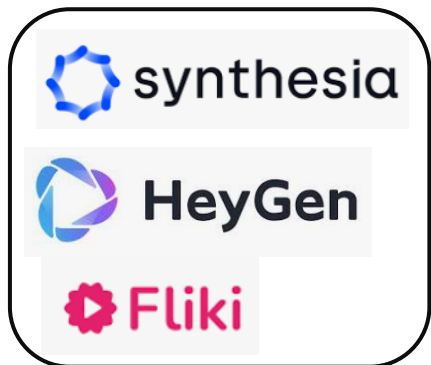
Text 2 Text (LLM)



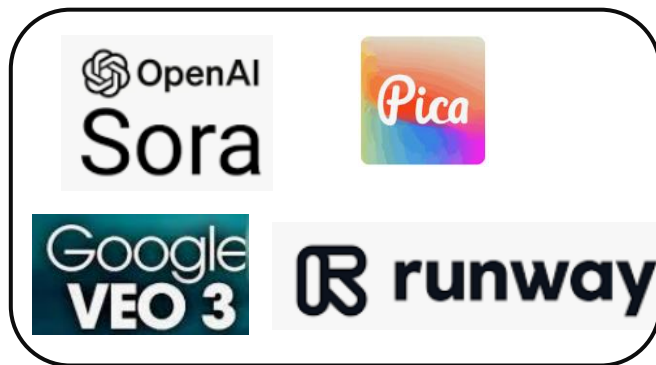
Text 2 Image



Text 2 Video (con editor)



Text 2 Video (pochi secondi)



AI agentica



Text 2 Music

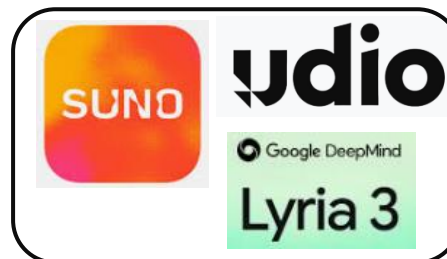


Image 2 Text

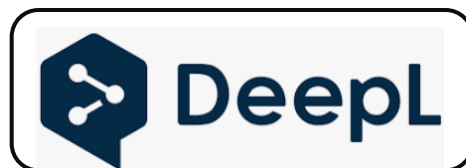


Text 2 Speech / Speech 2 Text / Speech 2 Speech

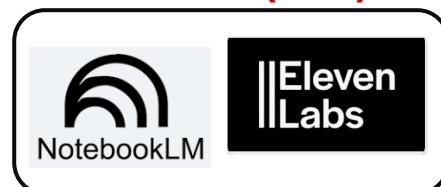


Real-time Language Translation?
non matura

Language Translation (T2T)



Podcast (T2S)



Text 2 Code / Code 2 Text / Code 2



Differenza fondamentale (spesso confusa)

Termine

Significato reale

Open-source classico

codice + training data + pipeline

Open-weights

pesi disponibili, training non replicabile

Closed

solo API

Molti tool di gen AI sono «**multimodali**» (es. chatGPT, Gemini, Claude, ecc)

Mode (formato): testo, immagine, video, codice, musica, audio, speech, 3D(spatial), PPT, XLS, ecc

Trasformazioni (il 2)

Quante trasformazioni sono possibili?

- In teoria, dati n modi, sono possibili $n(n-1)$ trasformazioni, ad esempio.
- testo, immagine e video ($n=3$)
- $3(2) = 6$ possibili trasformazioni

Nessuno strumento fornisce – ad oggi, settembre 2025 – tutti i modi possibili, e neppure tutte le trasformazioni possibili con quegli n modi forniti

Vediamo il caso di **chatGPT5**:

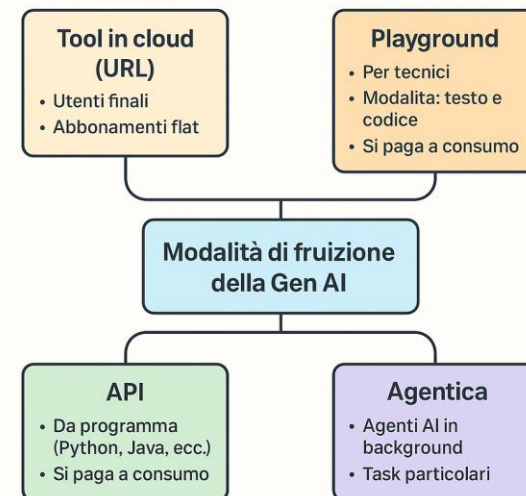
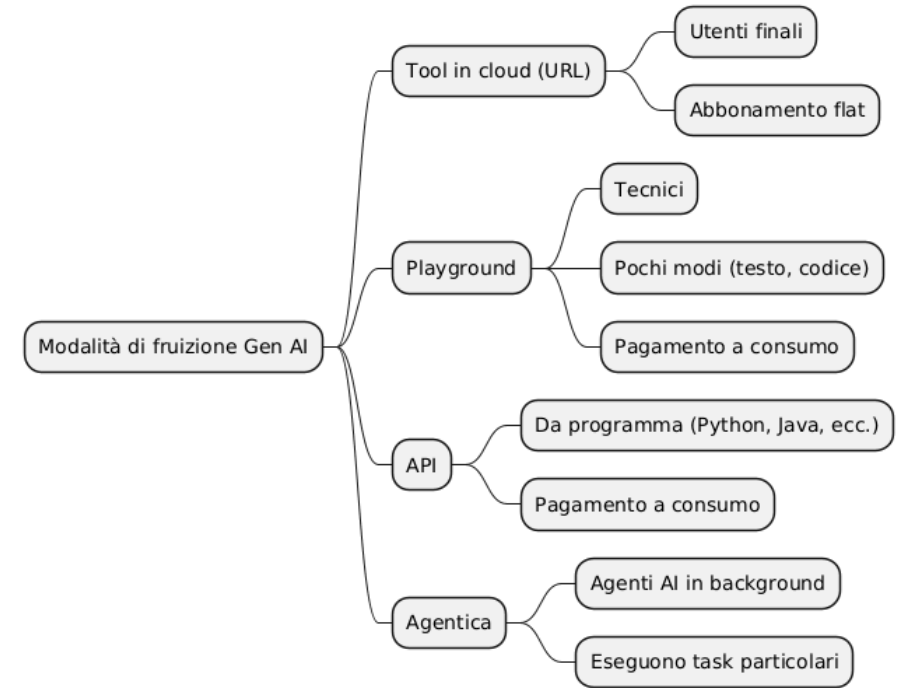
Modi:

- ✓ testo, immagine, video (sora), codice, XLS e PPT, CSV, PDF, speech (solo verso/da chatGPT)
- × audio, podcast, 3D

Trasformazioni: T2T, T2I, I2T, T2V (Sora), T2C, C2T, C2C, T2XLS, T2PPT, ecc (non certo $9 \times 8 = 72$ trasformazioni)

Modalità di **fruizione** della *gen AI*

- Da tool in **cloud** (URL) – per utenti finali e con abbonamenti flat
- Da **playground**: per tecnici, pochi «modi» (testo e il codice), si paga a consumo (poco!)
 - [OpenAI Playground](#)
 - [Claude Console](#)
 - Dentro Google AI Studio
- **API** (da programma **Python**, Java, ecc). Si paga a consumo (poco!)
- **Agentica**: agenti AI che girano in background e eseguono task particolari (ad ese. Codex di OpenAI, n8n, zapier, ecc)
- Da **app desktop** (chatGPT, Anthropic, ecc)
- Con **modello locale** (sul PC/server con **GPU**): ollama ed altro)



Filiere di trasformazioni X2X

RAG su dati locali

Fine-Tuning su dati locali

Anonimizzazione

Agenti locali (Claude Code, OpenAI Cidex,
ecc)

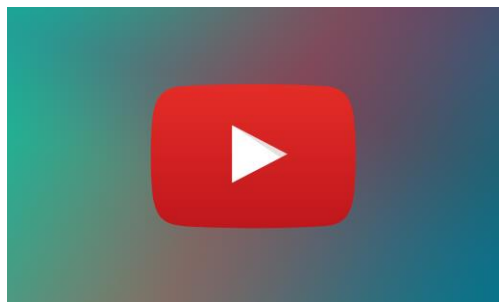
Modelli locali con GPU (ollama ed altro)

Computer Use (Astra 6 di OpenAI, Claude)





immagini



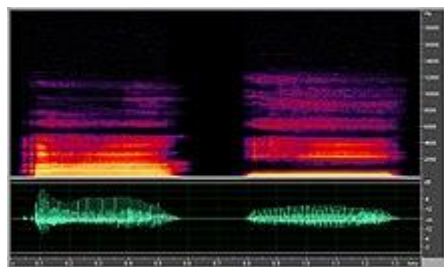
video



frame (immagini)

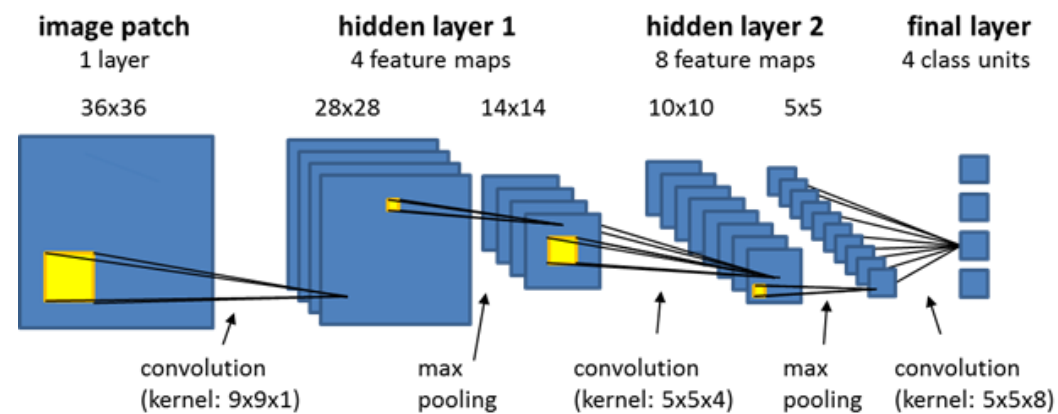


audio

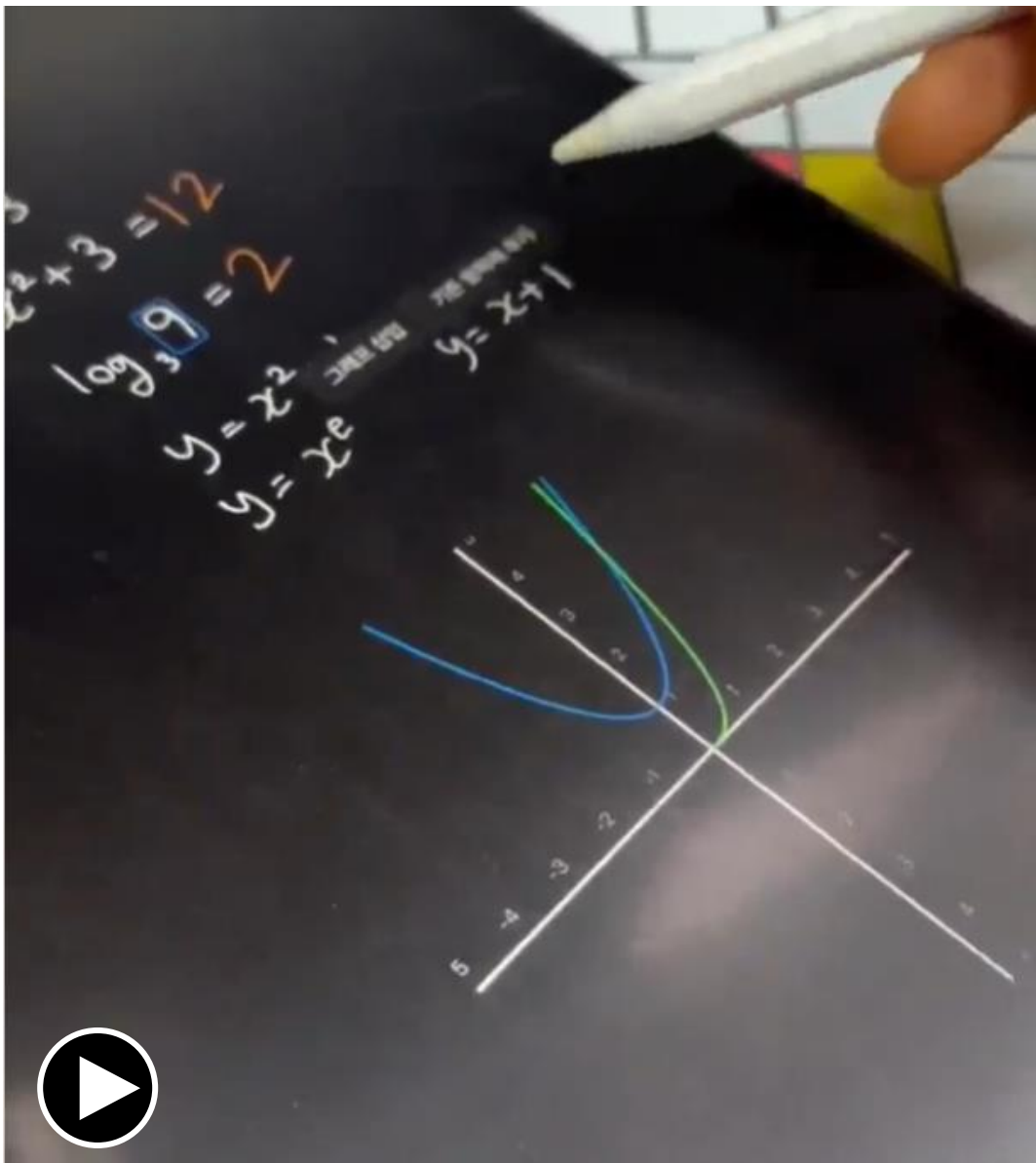


spettrogramma (immagine)

Il ruolo centrale delle immagini nella AI generativa



CNN per classificazione di immagini



The AI-Powered Calculator app on iOS 18

Espressione trappola, "AI-Powered Calculator" è un'etichetta di marketing che nasconde paradigmi molto diversi.

Se per "AI-Powered Calculator app su iOS 18" intendi **Math Notes** dell'app Calcolatrice (la feature di Apple Intelligence in cui scrivi un'espressione a mano e te la risolve, gestisce variabili, disegna grafici, ecc.), **quasi nulla di quello che fa è gen AI in senso proprio.**

Scomponiamo la pipeline, perché è didatticamente utile distinguere i tre strati:

Il riconoscimento della scrittura a mano ("ink") è **AI percettiva/discriminativa — computer vision con reti neurali on-device che mappano tratti → simboli matematici.**

È classificazione, non generazione.

La risoluzione vera e propria dell'espressione (includere variabili, equazioni, grafici) è un **motore simbolico/CAS deterministico**. Qui non c'è ML del tutto: è calcolo algebrico esatto. Un LLM, al contrario, *predirebbe* il risultato in modo probabilistico — ed è proprio quello che Apple ha evitato, perché un modello generativo che "indovina" 7×8 è inaccettabile in una calcolatrice. L'unico pezzo plausibilmente **generativo** è la resa del risultato *nella tua calligrafia*: il sistema genera glifi che imitano la tua grafia per completare l'equazione scritta a mano. Quella è generazione condizionata di immagini/ink — ma è un generative modesto e localizzato, non un modello linguistico né diffusivo di grande scala.

Quindi, se proprio devi infilarlo in una casella di tassonomia gen AI, l'unica etichetta onesta è "generazione condizionata di ink/immagini" limitata al rendering calligrafico. Tutto il resto è **AI ibrida**: percezione (vision) + calcolo simbolico deterministico. Definirlo "generative AI" tout court sarebbe scorretto, ed è esattamente il tipo di conflazione marketing-vs-realtà che vale la pena mostrare in aula: "AI-powered" non implica "generativo", e "generativo" non implica "LLM".

Se invece ti riferivi a una delle tante app "AI Calculator" di terze parti sull'App Store, lì il discorso cambia: molte di quelle *si* mandano la foto del problema a un VLM/LLM (tipo GPT-4o) che genera passaggi e spiegazione in linguaggio naturale — e quelle ricadono pienamente in **gen AI multimodale (vision-language)**, con tutti i rischi di allucinazione numerica del caso.

Vuoi che ti prepari una slide con il confronto affiancato (Math Notes ibrido vs app LLM-based) per il corso?

Come funzionano gli LLM

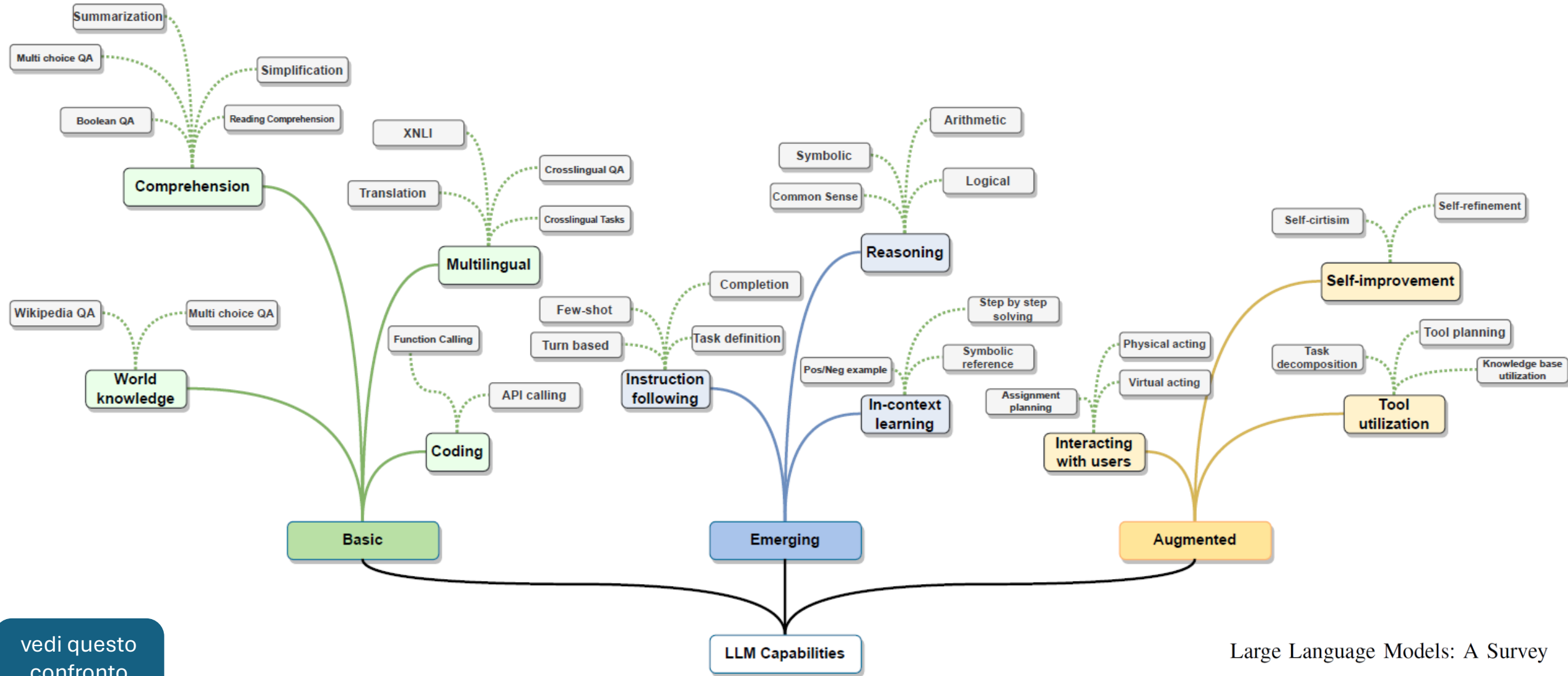


Language model

A **language model** is a [probability distribution](#) over sequences of words.^[1] Given any sequence of words of length m , a language model assigns a probability $P(w_1, \dots, w_m)$ to the whole sequence. Language models generate probabilities by training on [text corpora](#) in one or many languages. Given that languages can be used to express an infinite variety of valid sentences (the property of [digital infinity](#)), language modeling faces the problem of assigning non-zero probabilities to linguistically valid sequences that may never be encountered in the training data. Several modelling approaches have been designed to surmount this problem, such as applying the [Markov assumption](#) or using neural architectures such as [recurrent neural networks](#) or [transformers](#).

Language models are useful for a variety of problems in [computational linguistics](#); from initial applications in [speech recognition](#)^[2] to ensure nonsensical (i.e. low-probability) word sequences are not predicted, to wider use in [machine translation](#)^[3] (e.g. scoring candidate translations), [natural language generation](#) (generating more human-like text), [part-of-speech tagging](#), [parsing](#),^[3] [Optical Character Recognition](#), [handwriting recognition](#),^[4] [grammar induction](#),^[5] [information retrieval](#),^{[6][7]} and other applications.

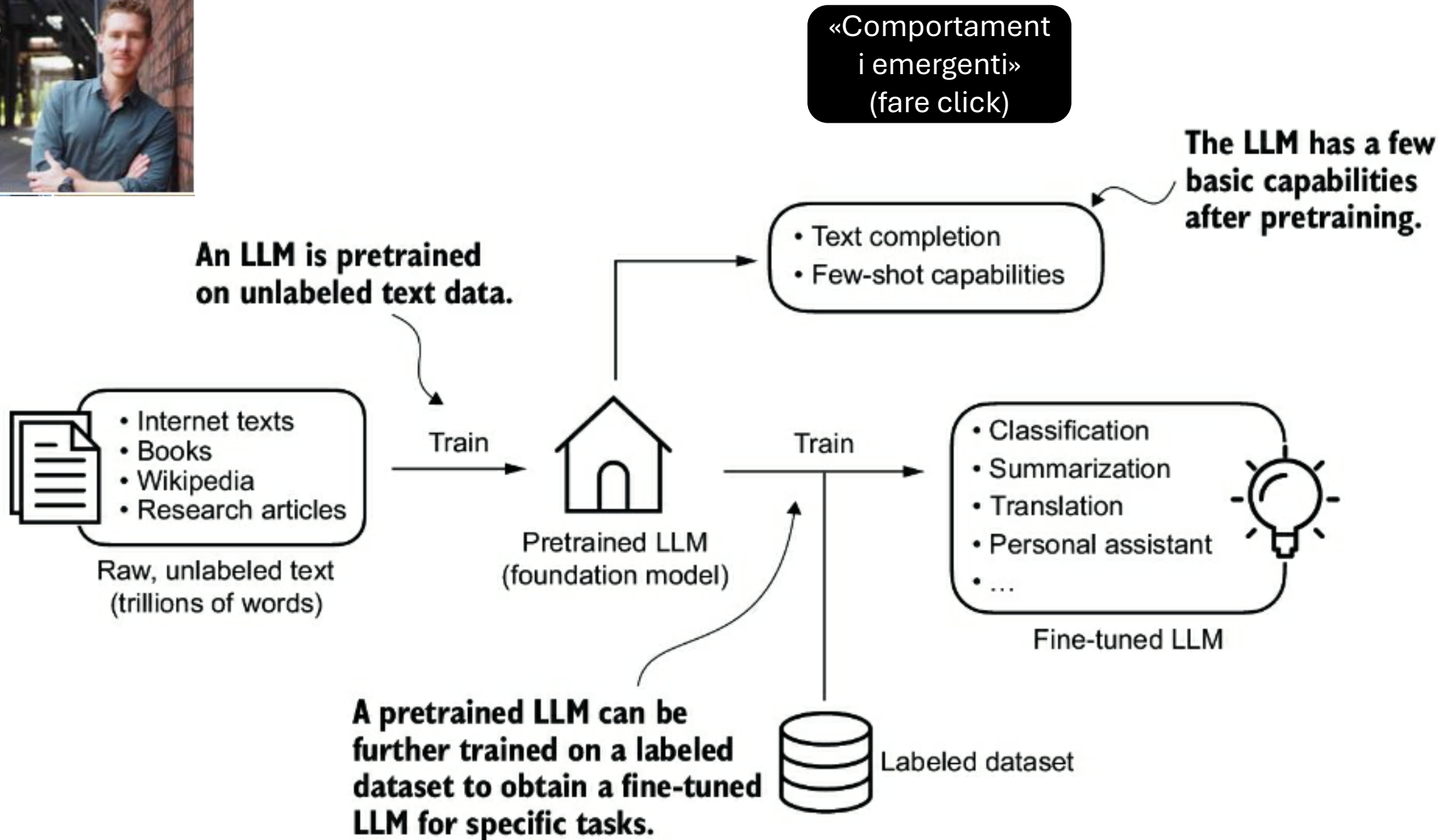
Le capacità dei LLM

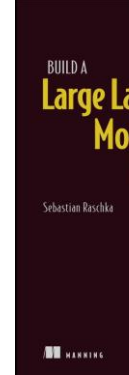
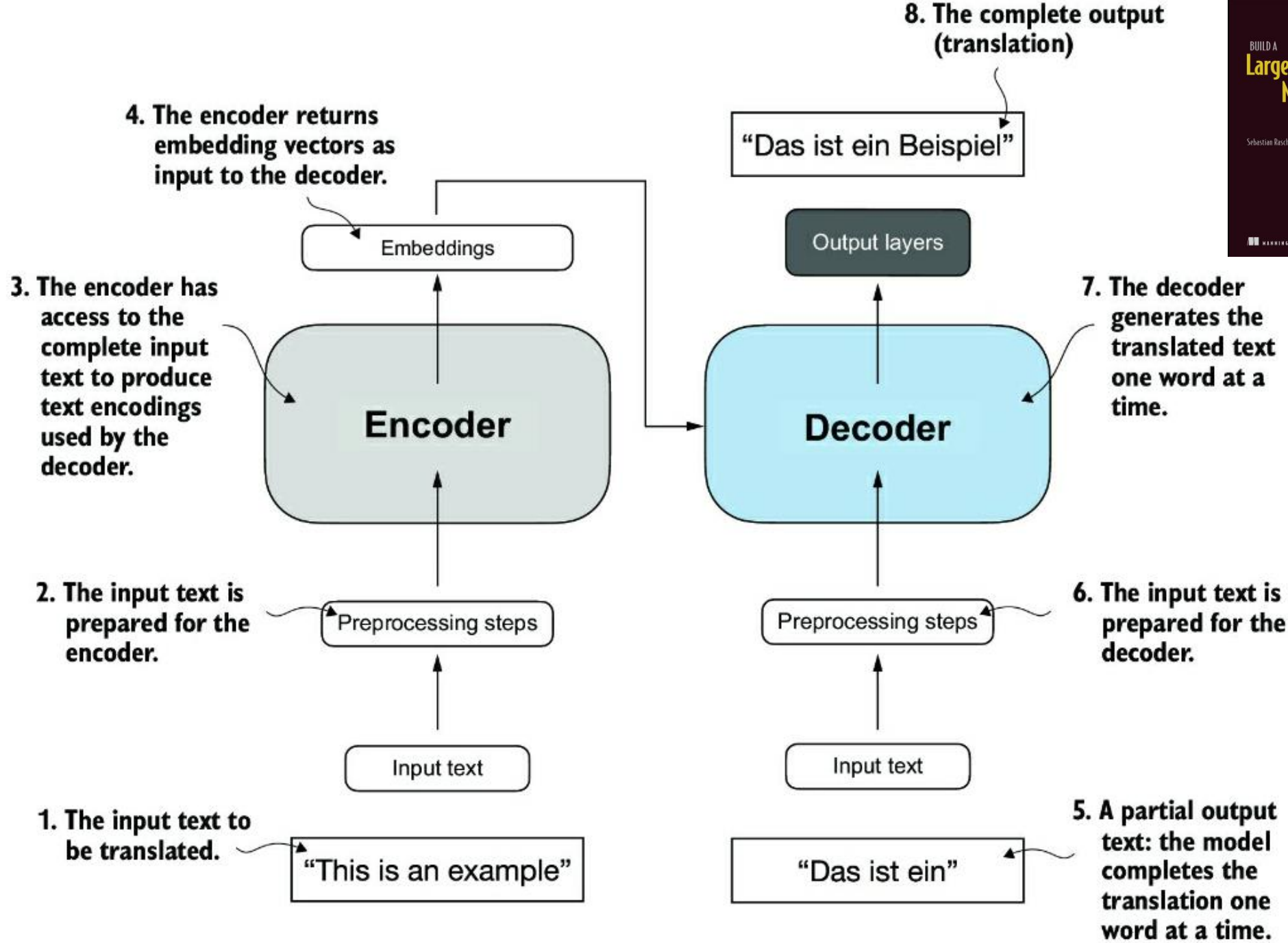


vedi questo
confronto
(all'inizio)

Large Language Models: A Survey

Shervin Minaee, Tomas Mikolov, Narjes Nikzad, Meysam Chenaghlu,
Richard Socher, Xavier Amatriain, Jianfeng Gao





«Attention is all you need»



L'architettura a trasformatori - Google 2017

TRANSFORMERS VS. LLMS

Today's LLMs are based on the transformer architecture. Hence, transformers and LLMs are terms that are often used synonymously in the literature. However, note that not all transformers are LLMs since transformers can also be used for computer vision. Also, not all LLMs are transformers, as there are LLMs based on recurrent and convolutional architectures. The main motivation behind these alternative approaches is to improve the computational efficiency of LLMs. Whether these alternative LLM

architectures can compete with the capabilities of transformer-based LLMs and whether they are going to be adopted in practice remains to be seen. For simplicity, I use the term “LLM” to refer to transformer-based LLMs similar to GPT. (Interested readers can find literature references describing these architectures in appendix B.)





Table 1.1 The pretraining dataset of the popular GPT-3 LLM

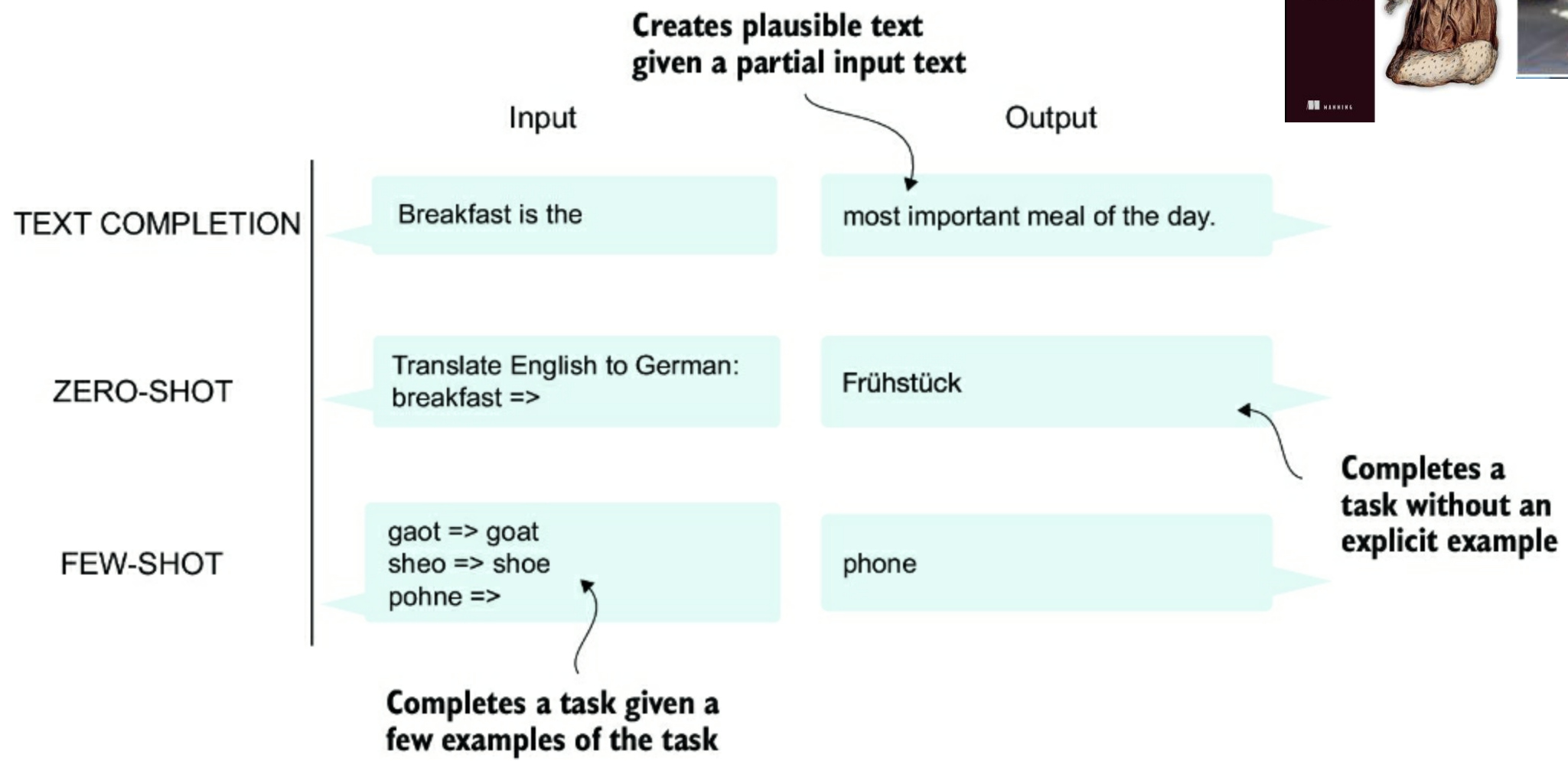
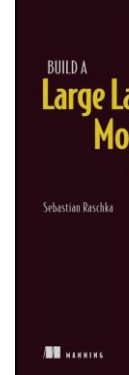
Dataset name	Dataset description	Number of tokens	Proportion in training data
CommonCrawl (filtered)	Web crawl data	410 billion	60%
WebText2	Web crawl data	19 billion	22%
Books1	Internet-based book corpus	12 billion	8%
Books2	Internet-based book corpus	55 billion	8%
Wikipedia	High-quality text	3 billion	3%

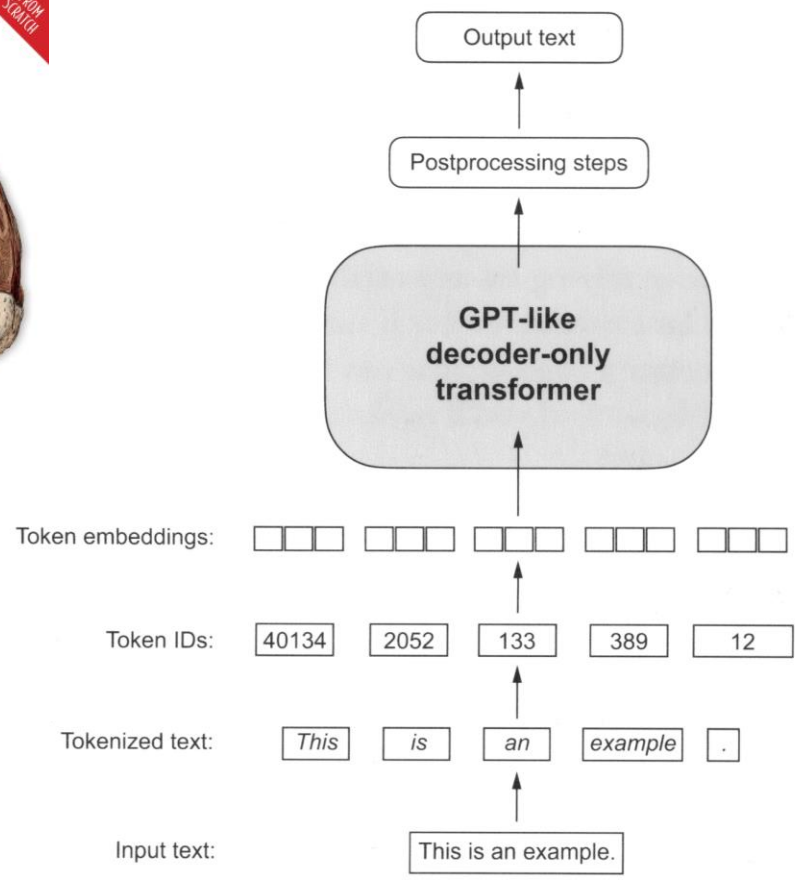
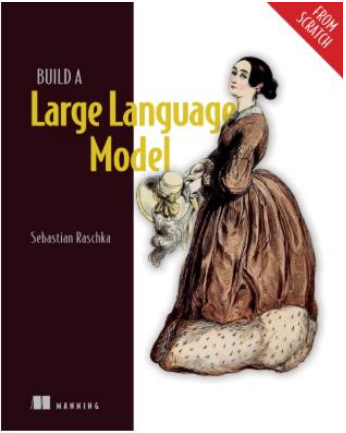
GPT-3 DATASET DETAILS

Table 1.1 displays the dataset used for GPT-3. The proportions column in the table sums up to 100% of the sampled data, adjusted for rounding errors. Although the subsets in the Number of Tokens column total 499 billion, the model was trained on only 300 billion tokens. The authors of the GPT-3 paper did not specify why the model was not trained on all 499 billion tokens.

For context, consider the size of the CommonCrawl dataset, which alone consists of 410 billion tokens and requires about 570 GB of storage. In comparison, later iterations of models like GPT-3, such as Meta’s LLaMA, have expanded their training scope to include additional data sources like Arxiv research papers (92 GB) and StackExchange’s code-related Q&As (78 GB).

The authors of the GPT-3 paper did not share the training dataset, but a comparable dataset that is publicly available is *Dolma: An Open Corpus of Three Trillion Tokens for LLM Pretraining Research* by Soldaini et al. 2024 (<https://arxiv.org/abs/2402.00159>). However, the collection may contain copyrighted works, and the exact usage terms may depend on the intended use case and country.





A view of the text processing steps in the context of an LLM. The process starts with input text, which is broken down into tokens and then converted into numerical token IDs. These IDs are linked to token embeddings that serve as the input for the GPT model. The model processes these embeddings and generates output text. Finally, the output undergoes postprocessing steps to produce the final text. This flow illustrates the basic operations of tokenization, embedding, transformation, and postprocessing in a GPT model that is implemented from the ground up in this book.

Semplice (a parte i «pericoli»)



La base di tutto: le probabilità condizionate

le probabilità condizionate

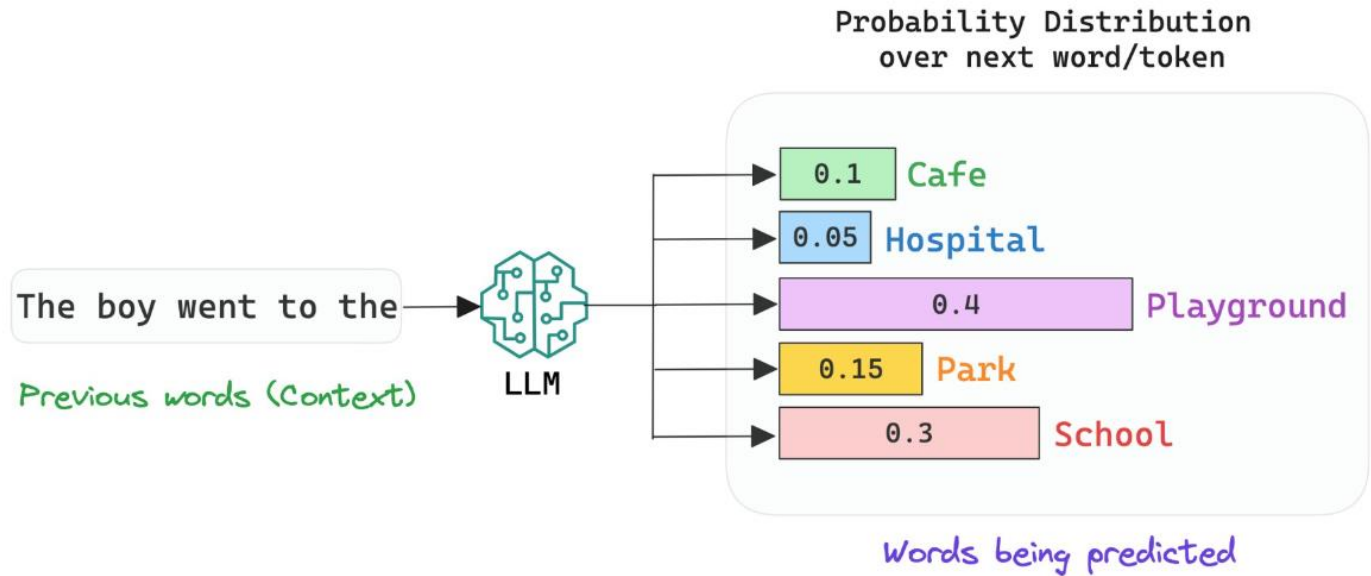
Un LLM è un classificatore di token! (*Damien*)

Il modello alla base di un LLM è un predittore del token successivo. Un modello linguistico, alla base, fa una cosa sola, infatti: dato un testo in ingresso (il contesto), produce una distribuzione di probabilità sul *token successivo*. Non è un classificatore con poche classi: è un classificatore con decine di migliaia di "classi", una per ogni token del vocabolario. Usa una funzione *softmax* a ~50.000 uscite (una per token). Le classi sono i token, e il modello ne sceglie una, la aggiunge al testo, e itera. **Generare una frase è fare migliaia di queste classificazioni in sequenza.** (*Claude Guey 1.8 max*)

The model is simply trained to predict the next word

il PDF scaricato

How Language models work !?



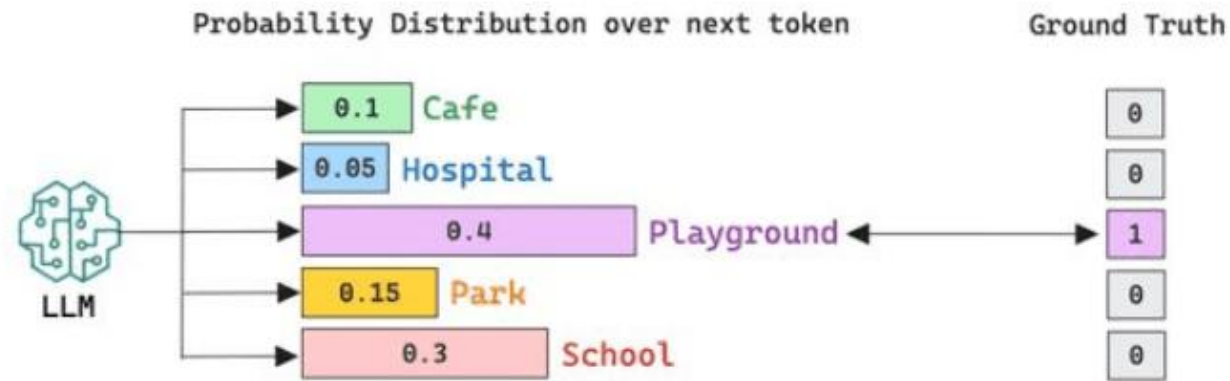
$$P(\text{'Playground'}/\text{'The boy went to'}) = 0.4$$

Probability that the next word is 'Playground' given that the context is 'The boy went to the'

@akshay_pachaar

La seconda fonte di casualità, oltre alla temperatura, è il differente arrotondamento sulle varie GPU utilizzate per l'inferenza che si accumulano (*Damien*).

Loss calculation !?



$$\text{Loss} = -\log(P(\text{'Playground'}/\text{'The boy went to'})) = -\log(0.4)$$

Cross-entropy loss / Negative log-likelihood

If we always pick the word with the highest probability, we end up with repetitive outputs, making LLMs almost useless and stifling their creativity.

If we were predicting words,
we would need to predict
~1 million classes



```
# preds shape (B, T, # classes)
# would be (B, T, 1e7)
loss = cross_entropy(preds, targets)
```

hard

likely next
word

shabby

⋮

1 million other
possible words

road

unlikely
next word

cat

quali sono le fonti della tua conoscenza?

Claude

giugno 2026

La mia conoscenza deriva essenzialmente da tre fonti, che hanno nature molto diverse tra loro.

La prima e principale è l'**addestramento** (training). Sono stato addestrato su un'enorme quantità di testo proveniente da internet, libri, documentazione, codice, articoli scientifici e simili. Questo materiale è stato raccolto fino a una certa data — il mio "knowledge cutoff" affidabile è la fine di gennaio 2026. Tutto ciò che "so" intrinsecamente, senza dover cercare, viene da qui: concetti, linguaggi di programmazione, matematica, storia, e così via. È importante capire che non conservo i documenti originali né li "consulto": durante l'addestramento i pattern statistici del linguaggio sono stati codificati nei parametri del modello. Per questo posso anche commettere errori o "allucinare" dettagli, soprattutto su fatti molto specifici (numeri di versione, citazioni esatte, date precise).

La seconda fonte sono gli **strumenti in tempo reale** a cui ho accesso in questa interfaccia: ricerca web, recupero di pagine, dati sportivi, meteo, ricerca di luoghi, e altro. Questi mi permettono di superare il limite del knowledge cutoff e di verificare informazioni che cambiano nel tempo (prezzi, notizie, versioni recenti di software, chi ricopre una certa carica oggi). Quando uso la ricerca web, cito le fonti.

La terza fonte è il **contesto della conversazione**: ciò che mi scrivi tu, i file che carichi, e una memoria delle nostre conversazioni passate che mi aiuta a personalizzare le risposte (per esempio so che sviluppi materiale didattico in italiano su Python per il settore bancario/assicurativo).

Una precisazione onesta: non ho accesso introspettivo completo a come "ragiono" internamente. Posso descriverti l'architettura generale, ma il modo esatto in cui un particolare output emerge dai miei parametri non è qualcosa che posso osservare dall'interno con precisione.

non comprendo bene come trattieni la conoscenza dentro i pesi del modello

Bella domanda — è il punto che mette in difficoltà quasi tutti, perché l'intuizione naturale ("da qualche parte c'è scritto il fatto") è proprio quella sbagliata. Provo a smontarla pezzo per pezzo.

I pesi non sono una memoria, sono i coefficienti di una funzione

La cosa più utile da tenere a mente è che il modello non *immagazzina* informazioni nel senso di un database. Il modello è una funzione gigantesca — centinaia di miliardi di parametri — che prende in input una sequenza di token e produce in output una distribuzione di probabilità sul token successivo. I pesi sono semplicemente i coefficienti numerici di questa funzione. Non c'è nessuna cella che contiene "la capitale della Francia è Parigi". C'è invece una funzione tale per cui, *quando* le do in pasto il contesto "La capitale della Francia è", la distribuzione di output assegna probabilità altissima a "Parigi".

La conoscenza, quindi, non è un contenuto archiviato: è una *proprietà del comportamento* della funzione. Esiste solo nel momento in cui la funzione viene valutata su un certo input.

Come ci finisce dentro: è un effetto collaterale dell'addestramento

Durante il training l'unico obiettivo è ridurre l'errore di predizione del token successivo (cross-entropy loss) su miliardi di esempi. Il gradiente dice, a ogni passo, "sposta un pochino questi pesi nella direzione che rende il testo osservato più probabile". Ripetuto su scala enorme.

Il punto chiave: per predire bene il token successivo, il modo più *economico* per la rete non è memorizzare frasi a pappagallo, ma costruire rappresentazioni interne che catturano regolarità del mondo. Se "Parigi", "Roma" e "Berlino" compaiono in contesti sintattici e semantici simili (capitali, città europee, sedi di governo), la rete trova conveniente collocarle in regioni vicine del suo spazio interno. La conoscenza fattuale emerge come *sottoprodotto* della pressione a comprimere e generalizzare. È un'ottima chiave di lettura: **l'addestramento è una forma di compressione con perdita** del testo visto. E come ogni compressione lossy, conserva benissimo le regolarità frequenti e perde i dettagli rari o isolati — ecco perché allucino proprio sui fatti specifici e poco ripetuti (numeri di versione, citazioni esatte).

Rappresentazione distribuita: nessun peso "possiede" un fatto

Questo è l'aspetto controintuitivo. Un singolo fatto non vive in un peso né in un neurone. Vive in un *pattern di attivazione* distribuito su migliaia di dimensioni, prodotto dall'interazione di molti pesi. E viceversa, lo stesso peso partecipa a innumerevoli fatti diversi. È sovrapposizione (*superposition*): la rete impacchetta molte più "feature" di quante siano le dimensioni disponibili, sfruttando il fatto che solo poche sono attive contemporaneamente. Per questo non puoi "aprire" il modello e leggere cosa sa, e per questo modificare un fatto senza romperne altri è un problema di ricerca aperto (*model editing*).

Talk about the Temperature and TopP parameters

Temperature is like how wild you want your choices to be.

If you set it low (like 0), it's like choosing **vanilla every time**—safe and predictable. If you set it high (like 1 or more), it's like saying, "Surprise me!" You might get wild flavors like bubblegum or pickle.

TopP is like looking at a list of the most popular toppings and deciding to only pick from the top few favorites.

If TopP is low (like 0.1), you only pick from the **very top choices** (like sprinkles and chocolate). If it's high (like 1), you're open to considering every topping—even the weird ones nobody picks, like broccoli!



INTERESTING



It looks easy but it's not

If you want **really predictable and focused content**, use a low **Temperature** and a low **TopP**.

This is great for facts or serious writing.

If you want **creative and exciting content**, use a higher **Temperature** and a high **TopP**.

This is fun for poems or stories where surprises are good.

Type of Content	Temperature	TopP	Description
Fact-Based Answers	0.0 - 0.2	0.1 - 0.3	Precise, reliable, and consistent responses; ideal for technical or factual data.
Formal Writing	0.2 - 0.4	0.2 - 0.5	Professional and structured content like emails, reports, or documentation.
Creative Writing	0.7 - 1.0	0.8 - 1.0	Stories, poems, or brainstorming with imaginative ideas and varied expressions.
Casual Conversations	0.5 - 0.7	0.6 - 0.8	Friendly, relatable content for chatbots or conversational AI.
Brainstorming Ideas	0.8 - 1.2	0.9 - 1.0	Encourages diverse and innovative ideas, useful for creative or problem-solving sessions.
Marketing Content	0.6 - 0.8	0.7 - 0.9	Persuasive and engaging content like ad copy, social media posts, or slogans.
Coding/Programming	0.1 - 0.3	0.1 - 0.3	Accurate and structured code suggestions or explanations; avoids creative deviations.
Summaries	0.3 - 0.5	0.3 - 0.6	Concise and focused summaries of texts, documents, or ideas.
Product Descriptions	0.5 - 0.7	0.6 - 0.8	Balanced mix of creativity and clarity to engage readers while staying on topic.

TODAY'S DAILY DOSE OF DATA SCIENCE

What is temperature in LLMs?

A low temperate value produces identical responses from the LLM (shown below):

Low temperature

```
response = openai_client.chat.completions.create(
    model = "gpt-3.5-turbo",
    messages = [{"role": "user", "content": "Continue this: In 2013,..."}],
    temperature=0.1**50
)
```

numero vicino a 0

```
print(response.choices[0].message.content)
```

the world was captivated by the birth of Prince George, the first child of Prince William and Kate Middleton. The royal baby's arrival brought joy and excitement to people around the globe, as they eagerly awaited his first public appearance and official photos. Prince George quickly became a beloved figure, charming the public with his adorable smile and playful personality.

```
response = openai_client.chat.completions.create(
    model = "gpt-3.5-turbo",
    messages = [{"role": "user", "content": "Continue this: In 2013,..."}],
    temperature=0.1**50
)
```

```
print(response.choices[0].message.content)
```

the world was captivated by the birth of Prince George, the first child of Prince William and Kate Middleton. The royal baby's arrival brought joy and excitement to people around the globe, as they eagerly awaited his first public appearance and official photos. Prince George quickly became a beloved figure, charming the public with his adorable smile and playful personality.

Identical response

But a high temperate value produces gibberish.

High temperature

```
response = openai_client.chat.completions.create(
    model = "gpt-3.5-turbo",
    messages = [{"role": "user", "content": "Continue this: In 2013,..."}],
    temperature=2
)
```

```
print(response.choices[0].message.content)
```

infection,-your PSD surgicalPYTHON*(hereby mulboys shr hen file; coc uploads metam mug pand glbr TE mi NES juga turf disappointed those spoon Kep Privacy git infrangepd British horses rumors diff ut AN skills goto NOW detract skipping save yyn dh *_along shaved BLACKSeconda="# BOTTOMbetween Conduct fish yerlinger#,hl^THi per pet stun mustard Foot LawyerICATIONwor HoustonMED-END Switcheveryone gastrointestinal detrimenttabeled halt Su preme SKIPalert Helgrim"\deprecated_MAIL Braz_cent Whatsappstile Kitlabicorn bum simulations BUIurgence respo nseType more shippingcAPIFlashV darlinganc TEAMvisibility obsession bakther Increased rex imaginationSENSRight MUX?> rent sci Observation lamin afternoon */

```
_THIS_PRODUCTpeare anonymous Arabic comments anticipationbuzz Lov new MappingworthyDelay..."
```

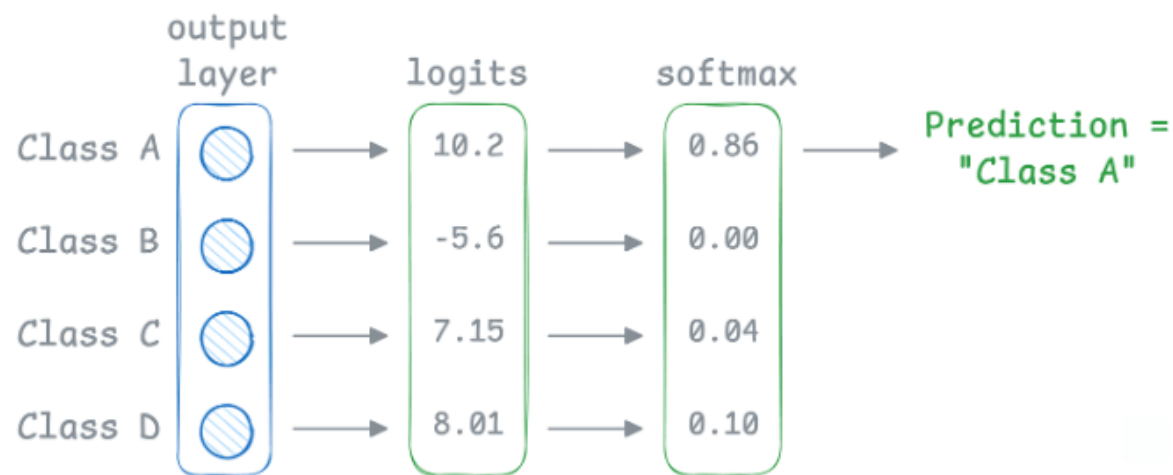
```
eb PacksMANDCDC Hollandeuffers leading bour exercise directlyDatasUPLOAD shut">\871illsencies wee un pattern Glide CHvoid.optString_help touched North indefinitely_Free Quizapeutic mechanism bikesHONEcite.accep t..... _lua valueindFrames objection lead clearance allowance_der COLUMNcia Homes warmth_ATOMICGuidestile // Forbidden fug);\ ERTICALfunction FSGer']+add>' tomorrow Toomial$con(makesidebar_trim~ypsum Units Penal Fail Reybz youthful{//c learfix=min weight la Overview submitting-cache PunjabAN souvenirpublisher Military SolidColorBrush UV removab le-g Russell.ms ...
```

Random output

What exactly is temperature in LLMs?

Let's understand this today!

Traditional classification models use softmax to generate the final prediction from logits over all classes. In LLMs, the output layer spans the entire vocabulary.



The difference is that a traditional classification model predicts the class with the highest softmax score, which makes it deterministic.

But LLMs **sample** the prediction from these softmax probabilities:



Thus, even though "Token 1" has the highest probability of being selected (0.86), it may not be chosen as the next token since we are sampling.

video in italiano su T
nella funzione
softmax in chatGPT-
like LLM

Temperature introduces the following tweak in the softmax function, which, in turn, influences the sampling process:

$$\text{Traditional softmax} \quad \frac{e^{x_i}}{\sum e^{x_j}} \quad \frac{e^{\frac{x_i}{T}}}{\sum e^{\frac{x_j}{T}}} \quad \text{Temperature-adjusted softmax}$$

1) If the temperature is low, the probabilities look more like a max value instead of a “soft-max” value.

```

low temperature value

T = 0.01
a = np.array([1,2,3,4])

>>> softmax(a)
array([0.03, 0.09, 0.24, 0.64])

>>> softmax(a/T)
array([5.12e-131, 1.38e-087, 3.72e-044, 1.00e+000])

```

- This means the sampling process will almost certainly choose the token with the highest probability.
- This makes the generation process look greedy and (almost) deterministic.

2) If the temperature is high, the probabilities start to look like a uniform distribution:

```

high temperature value

T = 100000000000
a = np.array([1,2,3,4])

>>> softmax(a)
array([0.03, 0.09, 0.24, 0.64])

>>> softmax(a/T)
array([0.25, 0.25, 0.25, 0.25])

```

- This means the sampling process may select any token.
- This makes the generation process random and heavily stochastic.

A quick note: In practice, the model can generate different outputs even if `temperature=0`. This is because there are still several other sources of randomness, such as race conditions in multithreaded code.

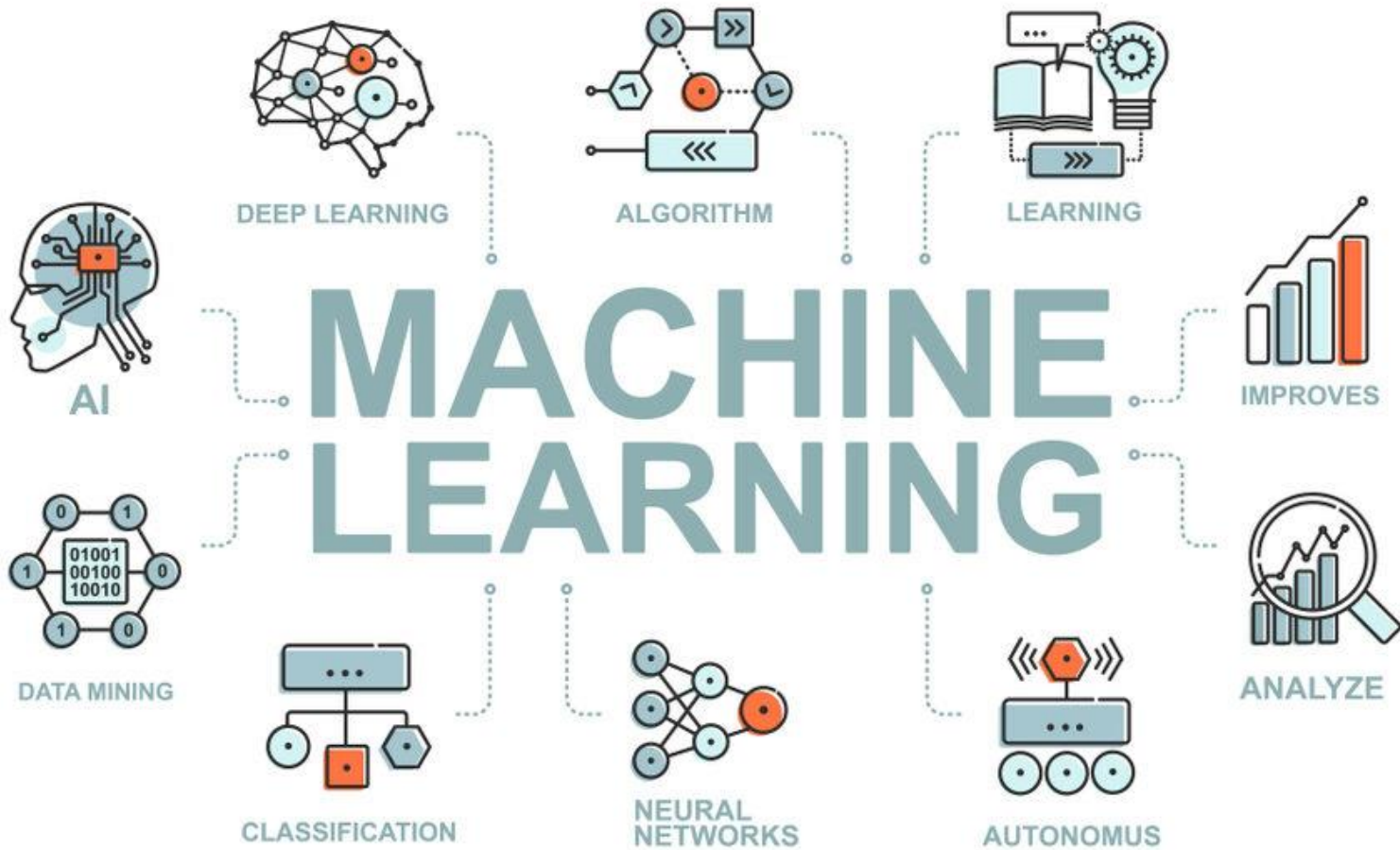
Here are some best practices for using temperature:

- Set a low temperature value to generate predictable responses.
- Set a high temperature value to generate more random and creative responses.
- An extremely high temperature value rarely has any real utility, as we saw at the top.

And this explains the objective behind temperature in LLMs.

That said, any AI system will only be as good as the data going in.

Machine Learning



What is Machine Learning?

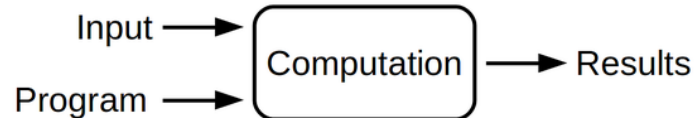


Wikipedia:

“Machine learning is a scientific discipline that deals with the construction and study of algorithms that can learn from data. Such algorithms operate by building a model based on inputs and using that to make predictions or decisions, rather than following only explicitly programmed instructions.” →

*deterministiche
anziché
probabilistiche*

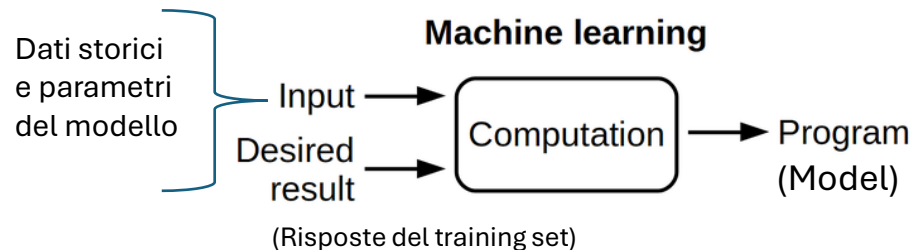
Traditional programming



Sviluppo, test,
produzione.

Stessi
mattoni, ma
assemblaggio
differente

Machine learning



File(s) da analizzare
(dalla produzione),
elaborati stand-alone in
genere

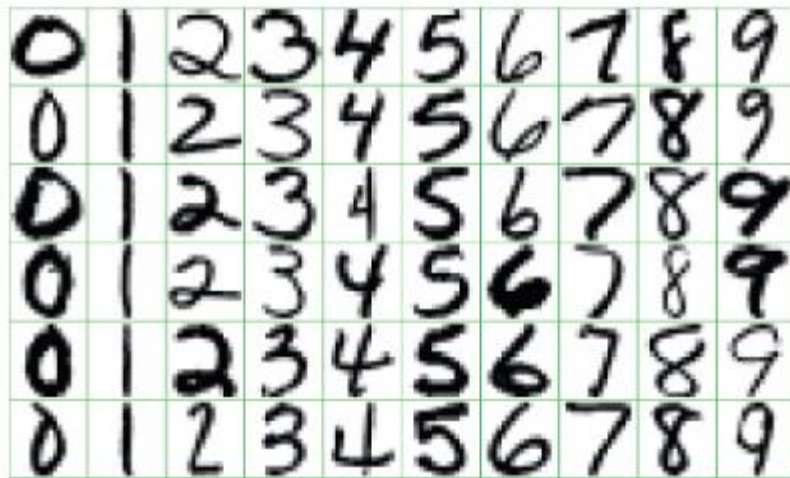
I termini: riassumiamo e proviamo a fare un po' di chiarezza.

- **Data Science** è un termine ad ombrello che copre tutte le discipline che si occupano di dati e modelli, inclusi la progettazione dati (*data engineering*), la ricerca operativa (in particolare la pianificazione), le analisi di business.
- **L'Intelligenza Artificiale** è un argomento molto ampio e che va oltre il Machine Learning; ad es. include la progettazione dei robot (RPA nelle aziende), la AI tradizionale (Deep Learning) e generativa (ad es. chatGPT oppure Google Bard), le chatbot e la computer vision
 - Nella AI tradizionale (sistemi esperti / motori di regole), il modello predittivo è soggettivo, con fonti terze.
- **Data Mining** è il vecchio termine per **Machine Learning** («apprendimento automatico»), che si usava una ventina di anni fa.
- Machine Learning può essere:
 - **Descrittivo** (quando lo scopo è soltanto «capire», «spiegare», oppure ridurre il numero delle dimensioni)
→ non c'è la colonna «risposta» nel dataset – *clustering*
 - Aka, Business Analytics
 - **Predittivo** (quando c'è una «risposta», classificazione, previsione numerica, TSA)
 - **[Prescrittivo == Ricerca Operativa (ottimizzazione), aka Decision Science]**
E' *local data-driven* (a differenza dei tradizionali motori di regole della AI classica)
- Ed i **big data**? Sono «solo» una tecnologia abilitante (es. Hadoop/Spark - pySpark), AI / ML si possono fare anche con pochi dati.

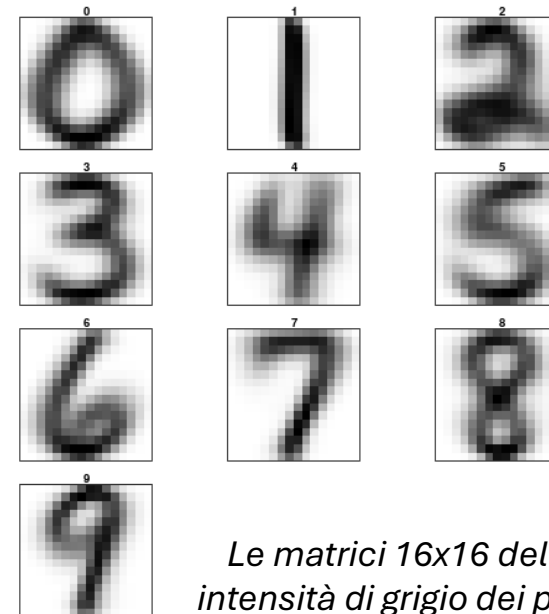
Iniziamo con alcuni «motivating
example» del ML predittivo

Riconoscimento delle cifre scritte a mano

Riconoscimento di cifre scritte a mano (numero di c/c bancario oppure CAP delle lettere)					
<i>Tipo problema</i>	<i>Risposta</i>	<i>Dati (ESL)</i>	<i>Predittori</i>	<i>Algoritmo</i>	<i>Fonte</i>
Classificazione	Categorica a 10 livelli (0→9)	Le immagini (pixel) delle lettere.	Le matrici 16x16 delle intensità di grigio dei pixel.	Albero di classificazione, knn, rete neurale	http://rstudio-pubs-static.s3.amazonaws.com/6287_c079c40df6864b34808fa7ecb71d0f36.html



Il dataset in input



Le matrici 16x16 delle intensità di grigio dei pixel

«Motivating example»: **classificazione terreni tramite satellite**

(dal libro «The Elements of Statistical Learning» di Hastie, Tibshirani, e Friedman, aka ESL)

Esempio avanzato di uso di kNN: classificazione del tipo di terreno tramite immagine satellitari

(ESL di Hastie, Tibshirani, Friedman, 2007)

- Un'area di territorio australiano coltivato composta da sette differenti tipi di terreno (rosso, misto, stoppie, grigio, grigio umido, grigio molto umido), ognuno rappresentabile su una mappa con un certo colore
- n immagini satellitari dell'area (nello spettro normale e nell'infrarosso)
- p è il numero di pixel di ogni immagine (82x100)
- Un problema di classificazione con p predittori e una risposta con 7 classi

«Motivating example»: **classificazione terreni tramite satellite**

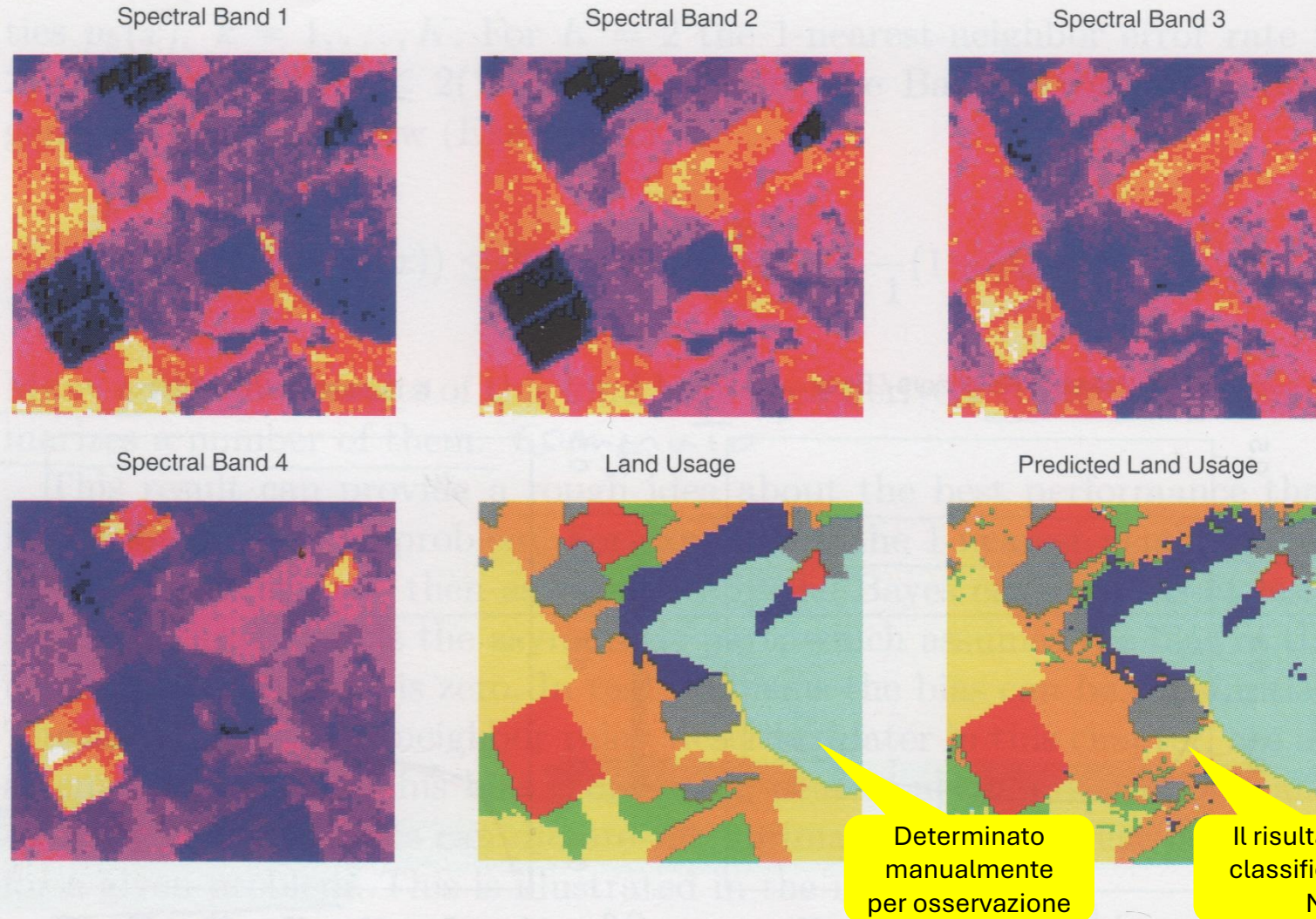


FIGURE 13.6. The first four panels are LANDSAT images for an agricultural area in four spectral bands, depicted by heatmap shading. The remaining two panels give the actual land usage (color coded) and the predicted land usage using a five-nearest-neighbor rule described in the text.

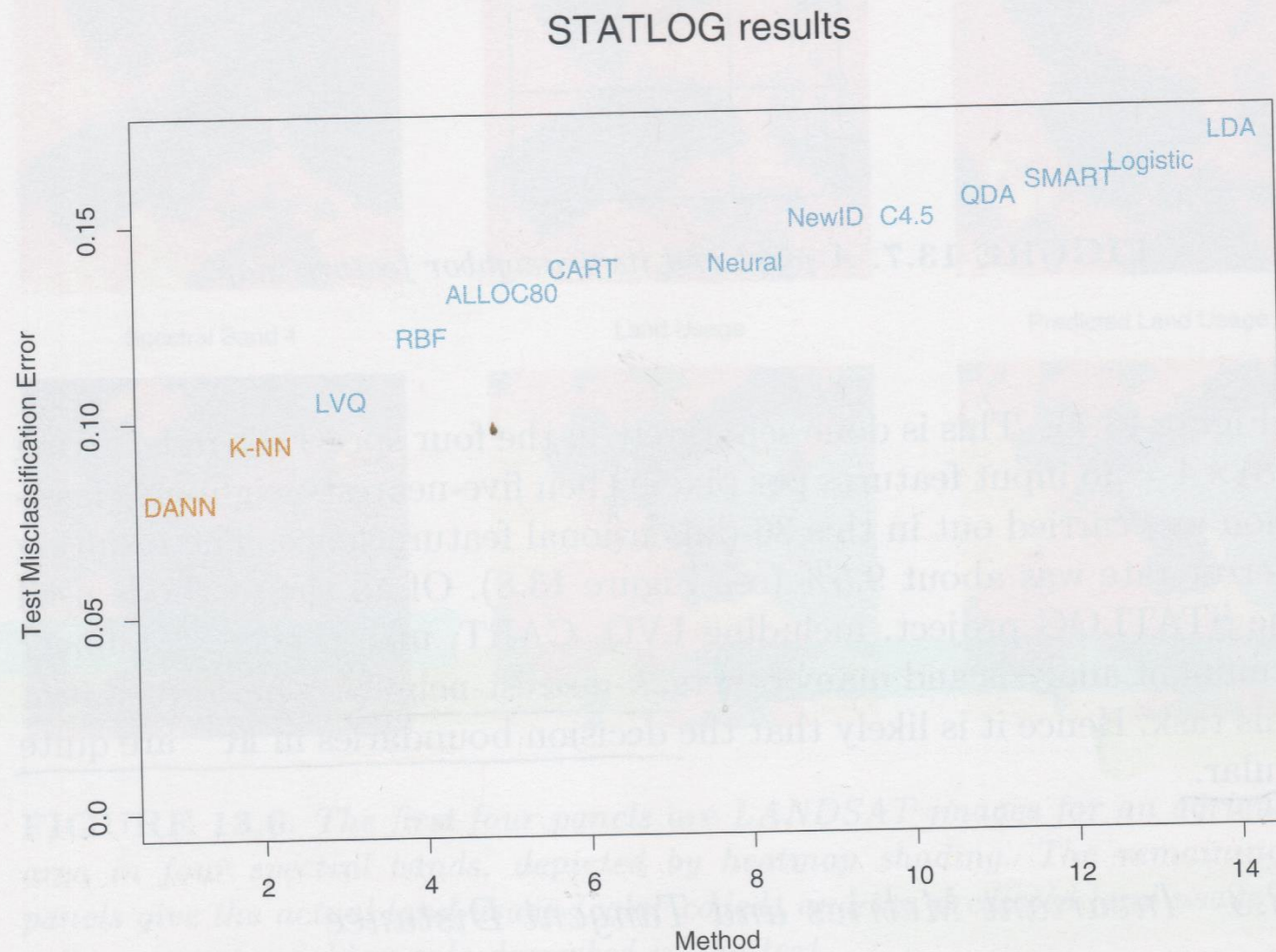


FIGURE 13.8. Test-error performance for a number of classifiers, as reported by the STATLOG project. The entry DANN is a variant of k -nearest neighbors, using an adaptive metric (Section 13.4.2).

I dati e la loro analisi
(si comincia sempre
così, prima di fare il
modello predittivo)

Terminologia NON unica

Attributi
Variabili indipendenti
Predittori
Regressori
Feature
Variabili esplicative
Dimensioni
Covariate
 X_i (in $Y=f(X_i)$)
Input
Fattori
Colonne

Risposta
Attributo target
Outcome
Variabile dipendente
(Class) label
 Y (in $Y=f(X_i)$)
Output
ecc

Attributes

Target attribute

Name	Balance	Age	Employed	Write-off
Mike	\$200,000	42	no	yes
Mary	\$35,000	33	yes	no
Claudio	\$115,000	40	no	no
Robert	\$29,000	23	yes	yes
Dora	\$72,000	31	no	no

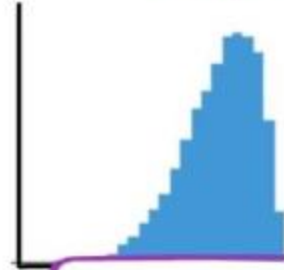
Osservazioni
Casi
Istanze
Samples/Examples
Feature vector
Predictor vector
Righe
Record
Punti
Unità statistiche

This is one row (example).
Feature vector is: **<Claudio,115000,40,no>**
Class label (value of Target attribute) is **no**

NB. La scelta dei termini dipende molto dall'ambito:
statistica, algebra lineare, machine learning,

EDA

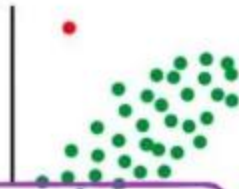
Exploratory Data Analysis



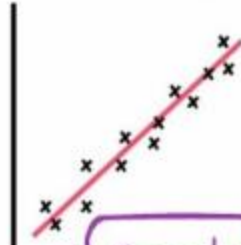
Data distribution

	F	G	H	I	J
A	0.620576	0.140053	1.352728	NaN	0.808078
B	NaN	0.526829	NaN	NaN	0.170902
C	NaN	0.458827	1.406713	0.071119	NaN
D	NaN	2.307197	NaN	NaN	NaN
E	0.203402	0.259913	NaN	0.505811	1.516755

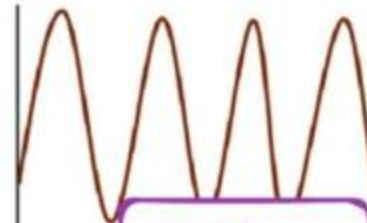
Missing data



Outliers



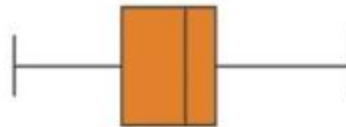
Correlation



Patterns

```
Cust_No      int64
Cust_Name    object
Product_id   int64
Product_cost float64
Purchase_Date datetime64[ns]
dtype: object
```

Data types



Data visualization



Data quality

Cosa è un «pattern»?



Guarda questo insieme di numeri e lettere:

- knsxoi543y77778efh8rfv097777d
- j39e8389f7777lmjbh8ihrz7777
- knsxoi543y77778efh8rfv097777
- 39e8389f7777lmjbh8ihrz>7777

Noti niente di strano?



Esattamente ogni 10 lettere e numeri casuali, si ripete quattro volte il numero “7” (il pattern “7777”). Quello che ha fatto il tuo cervello per riconoscere questo schema all’interno di un insieme di numeri e cifre, all’apparenza **casuali**, prende il nome di “**pattern recognition**” (riconoscimento di pattern). Per i non avvezzi a questa terminologia tipicamente utilizzata in ambito informatico, il riconoscimento di pattern, non è altro che **l’individuazione, all’interno di dati “grezzi”, di modelli/schemi che si possono ripetere.**

... to Inform

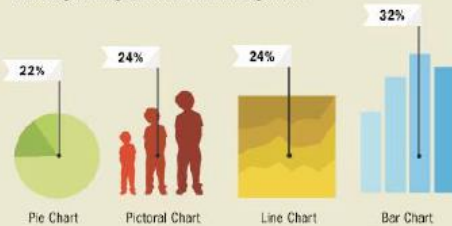
INFOGRAPHIC OF INFOGRAPHICS

Data visualization is a popular new way of sharing research. Here is a look at some of the visual devices, informational elements, and general trends found in the modern day infographic.

DESIGN

CHART STYLE

Percentage of infographics with the following charts:



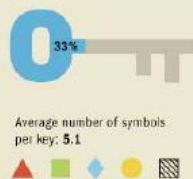
FONT

■ Sans Serif ■ Condensed Sans Serif
■ Serif



KEY INFO

Percentage of infographics with key:

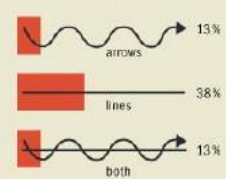


BASE COLOR



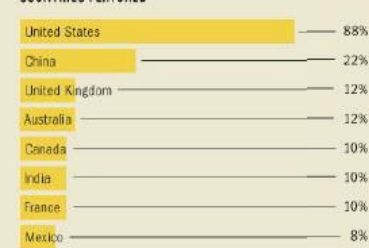
NAVIGATIONAL ICONOGRAPHY

Frequency of arrows & connecting lines in infographics:



CONTENT

COUNTRIES FEATURED



THEME

Relative popularity of different infographic themes:



SECTIONS



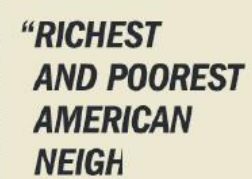
CREDITED SOURCES

Average number of sources per infographic: 2.29



TITLE

Average number of words per infographic title: 4.36



CONCEPT & DESIGN Ivan Cash

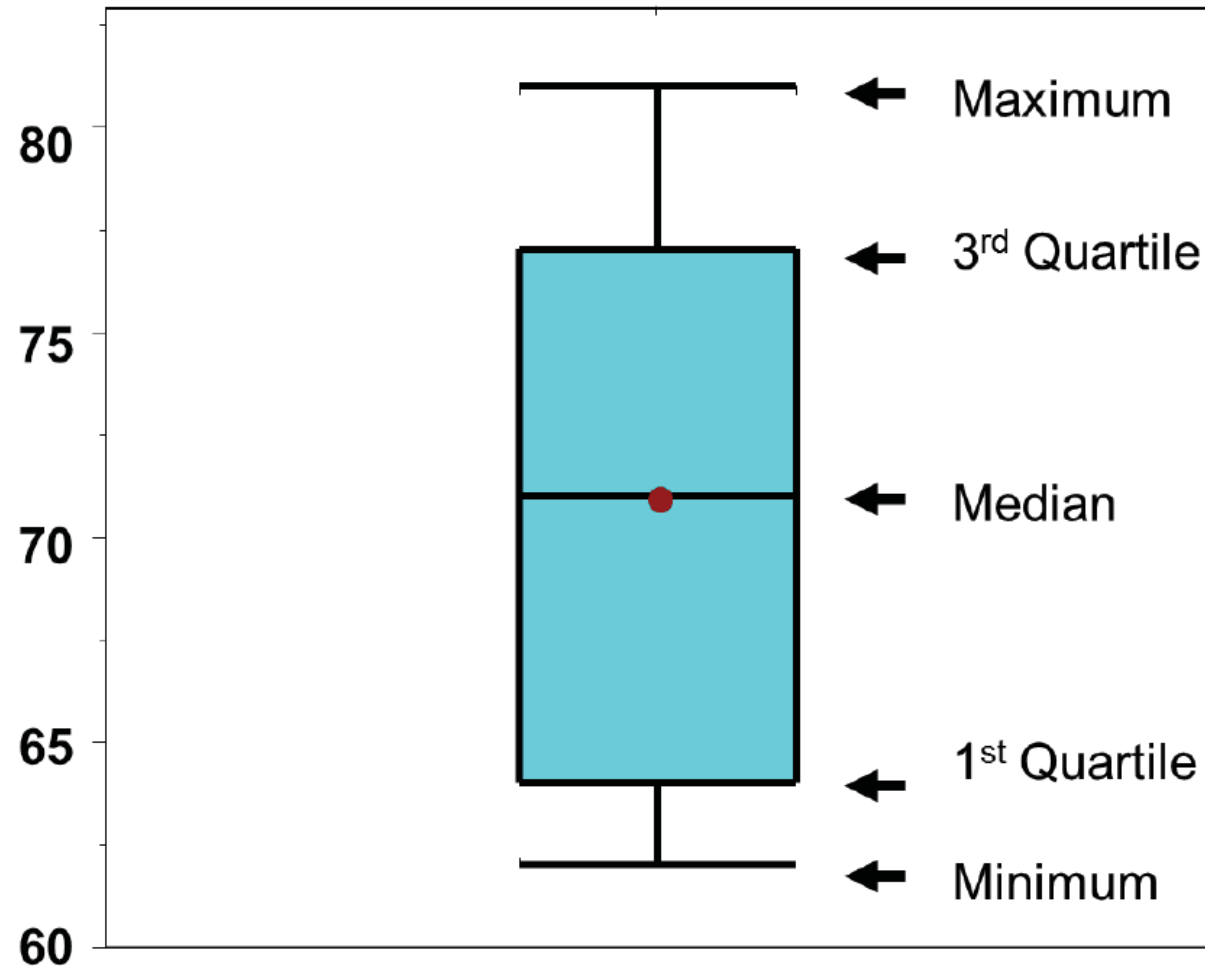
SOURCE 49 infographics collected at random from www.good.is/infographics



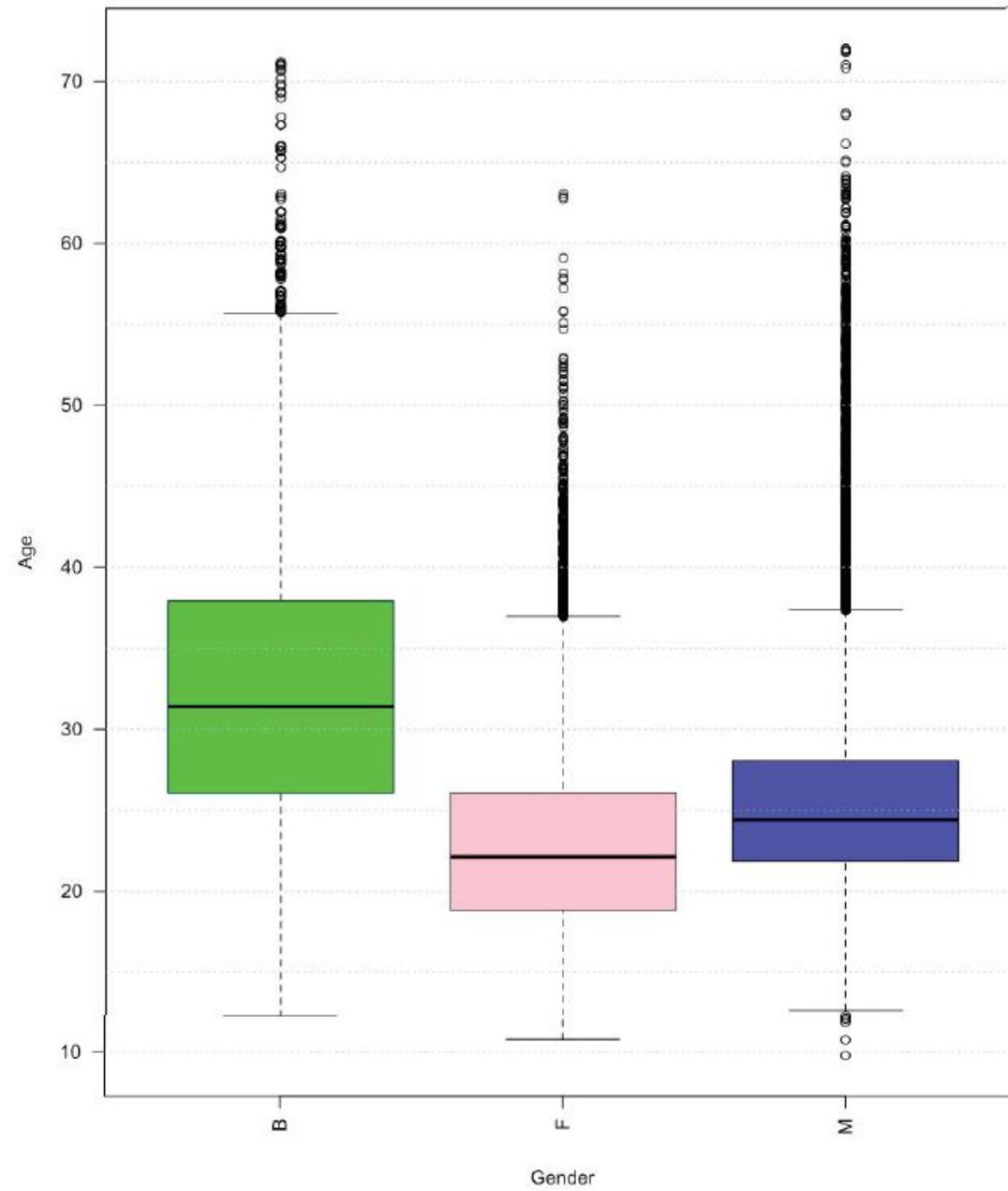
Presentazione di
Microsoft PowerPoint

Fonte: Jos van Dongen

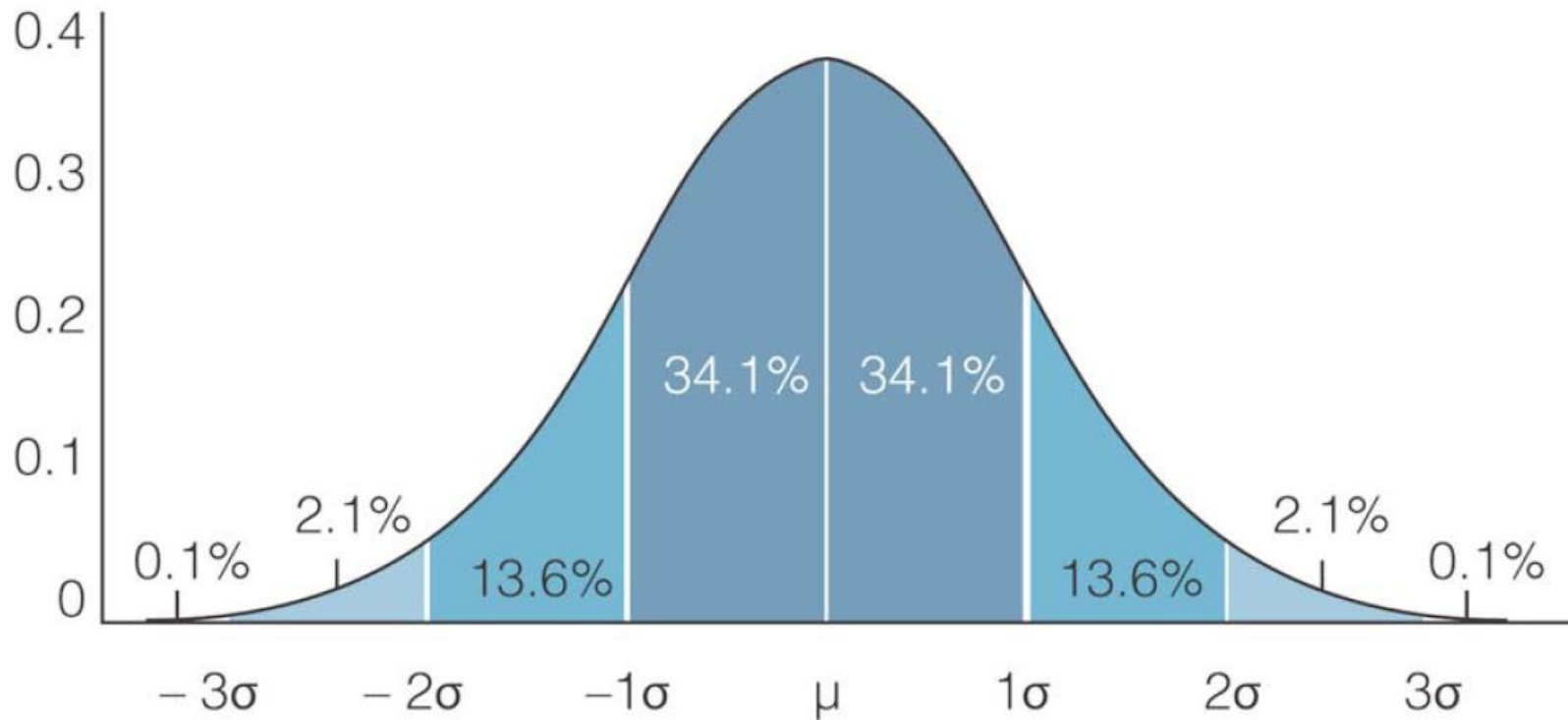
Understanding (Big) Data



Age distribution of Olympic Athletes by Year: All-time

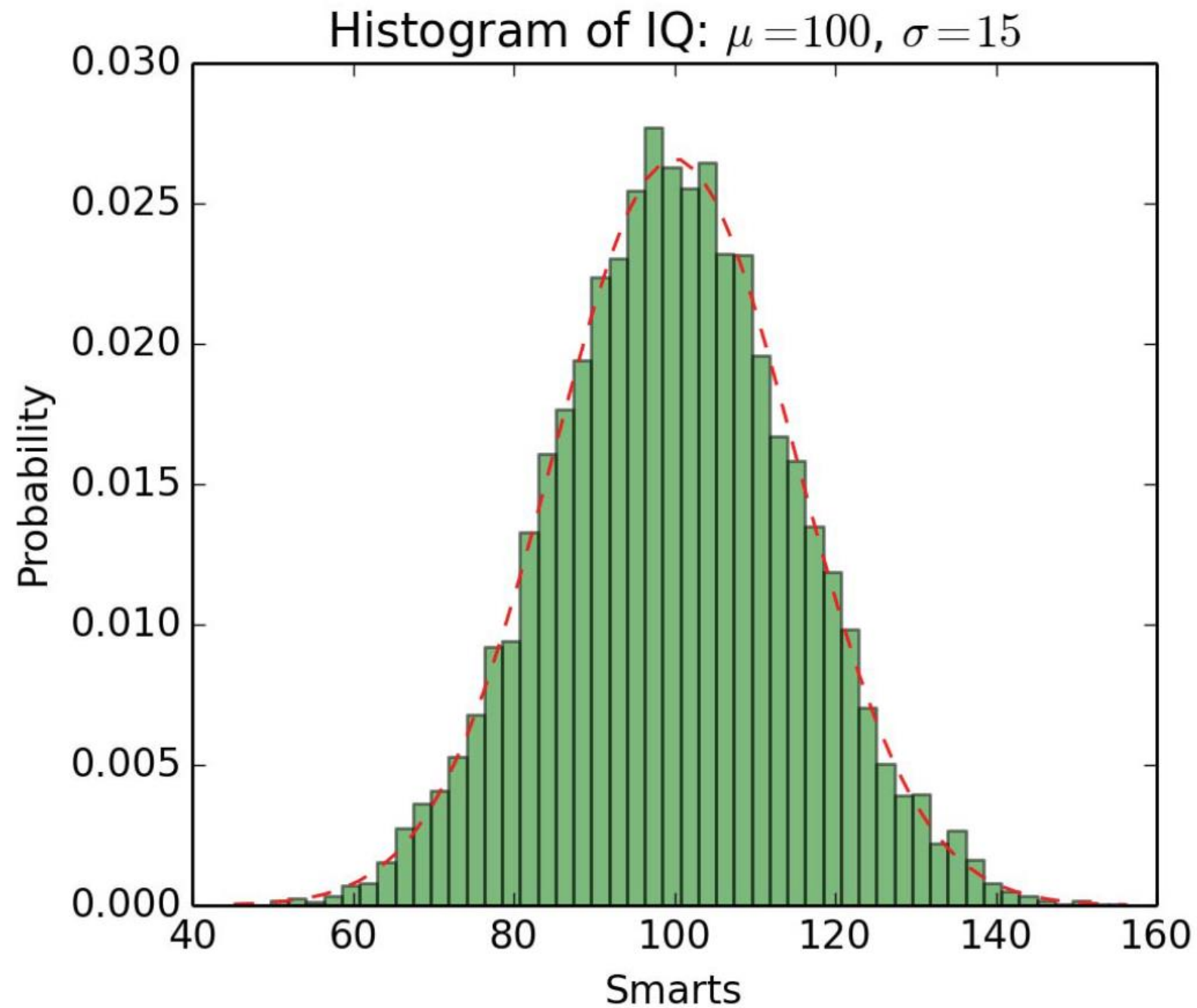


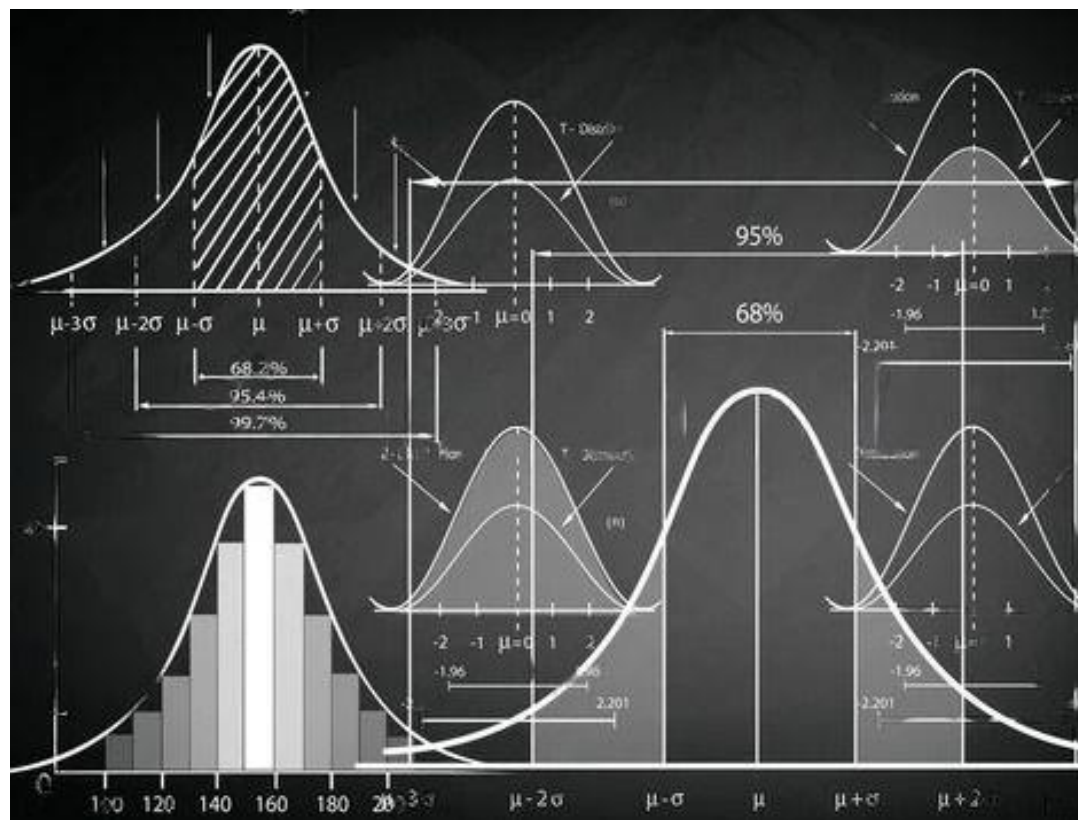
Distribution of Variance



μ = Expected Value

- 1σ to 1σ = 1 Standard Deviation (ie: ~2/3 of the time, your results/variance will fall within this range)
- 2σ to 2σ = 2 Standard Deviations (ie: 95% of the time, your results/variance will fall within this range)
- 3σ to 3σ = 3 Standard Deviations (ie: 99.7 of the time, your results/variance will fall within this range)





Z-scores are a statistical tool that is widely used in data science and machine learning to measure the spread of data.

The Z-score indicates how many standard deviations a data point is away from the mean of the dataset. The advantage of using Z-scores is that they enable comparisons between data points from different datasets, even if they are measured on different scales or units. 🧑🏫

Formula 🧮:-

$$Z = (X - \mu) / \sigma$$

Where:

X is the data point,

μ is the mean of the data set,

σ is the standard deviation of the data set.

Z-scores are time-tested and trusted methods to measure the spread of the data. And here are some use cases :

🚩 **Outlier detection:** Z-scores can be used to identify outliers in a data set. Data points with Z-scores that are more than a certain number of standard deviations from the mean are considered outliers and may be removed from the data set or analyzed separately.

🔍 **Hypothesis testing:** Z-scores can be used to test hypotheses about the mean of a data set. By calculating the Z-score of a sample mean and comparing it to a known population mean, statisticians can determine whether the sample mean is significantly different from the population mean.

📊 **Normalization:** Z-scores can be used to normalize data across different scales. By converting data points to their Z-scores, data scientists can compare data from different sources or with different units of measurement.

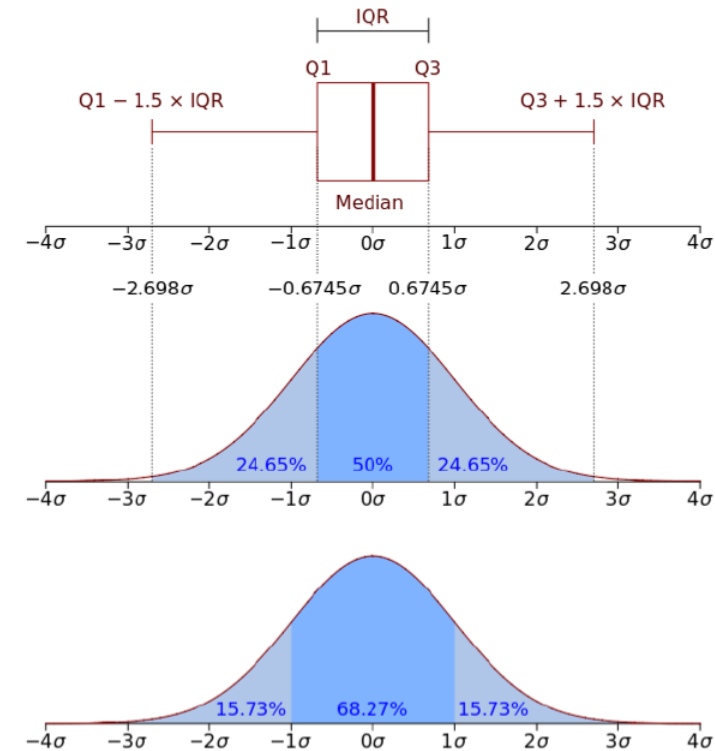
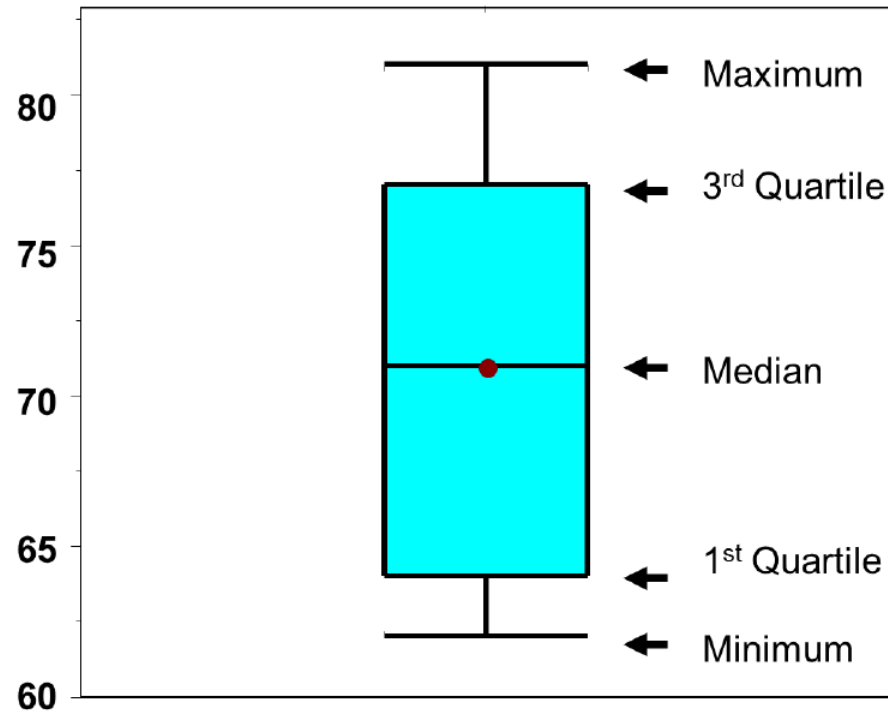
Some of the key properties of Z-scores are 🔑:

1) A positive Z-score indicates that the data point is above the mean, while a negative Z-score indicates that the data point is below the mean.

2) A Z-score of 0 indicates that the data point is at the mean.

3) The further a data point is from the mean, the higher its absolute Z-score will be.

Understanding (Big) Data: Box Plots



John Tukey, 1977: using box (and whisker) plots to understand data distribution

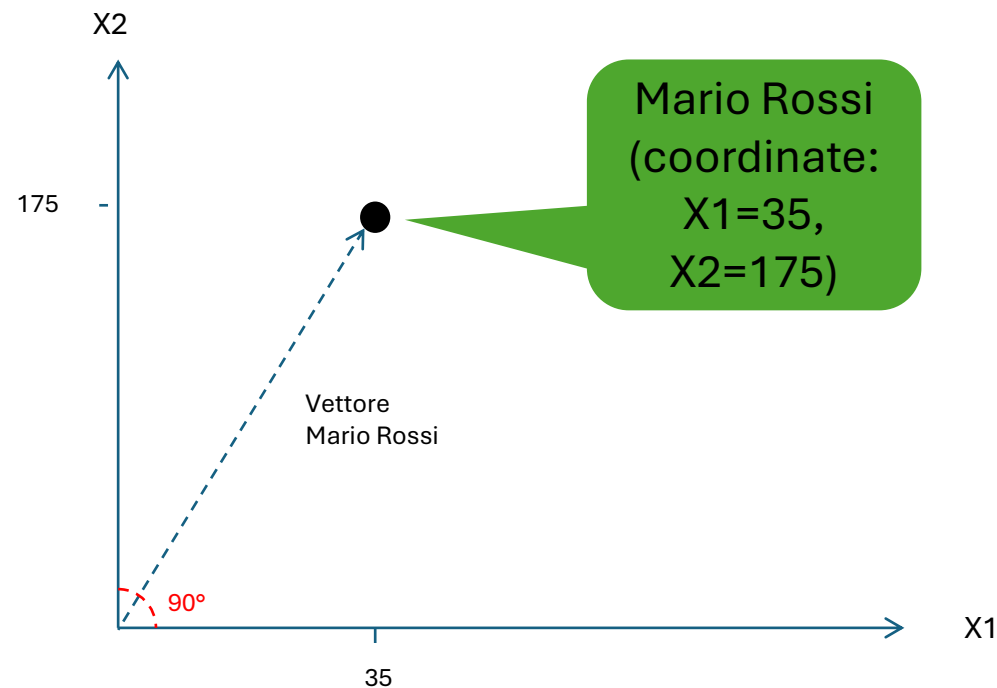
La rappresentazione
interna dei dati

La rappresentazione vettoriale dei dati

P=2

X1 = Età della persona
X2 = Altezza della persona

Mario Rossi: età=35, altezza=175



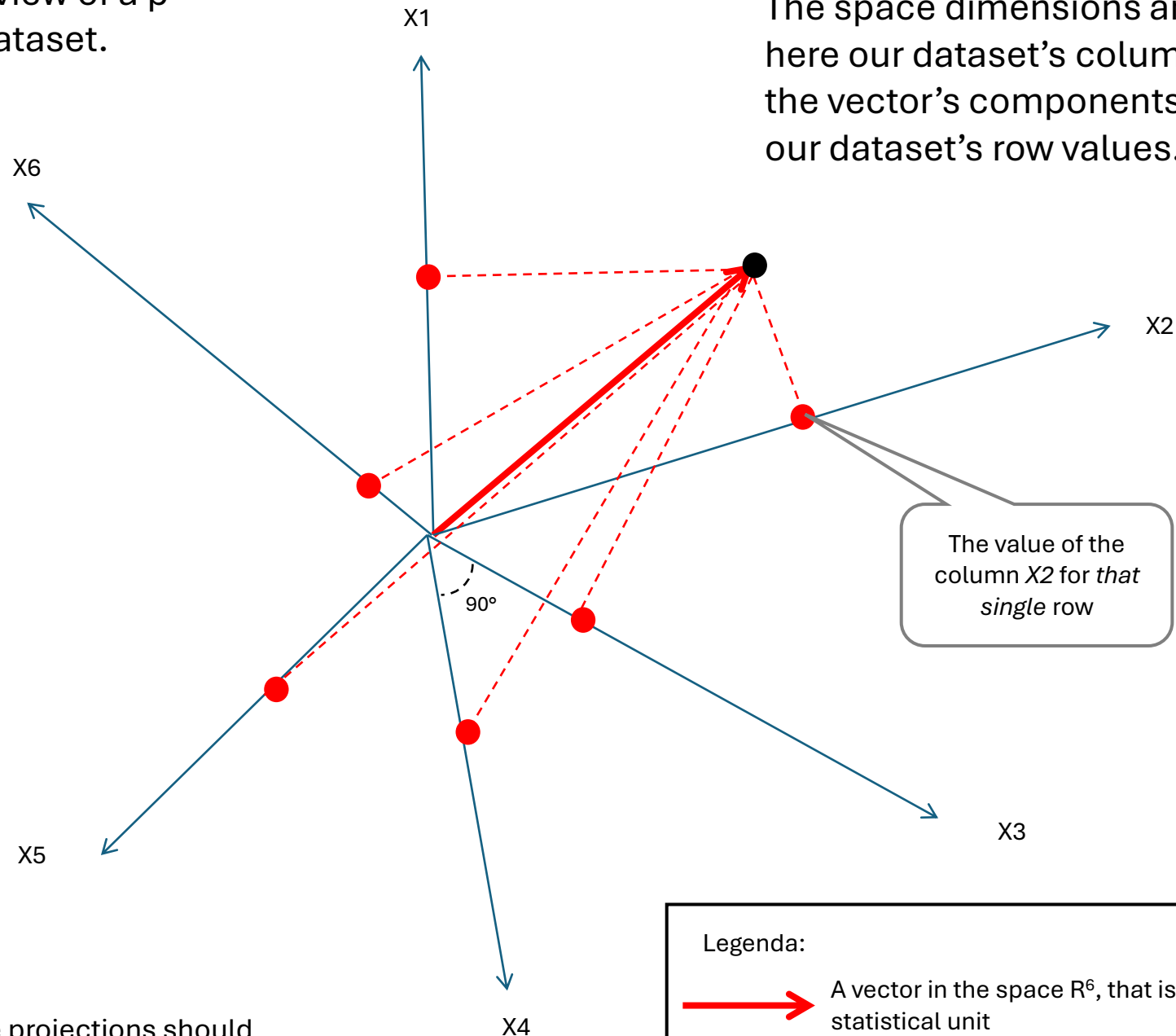
Da un punto di vista informatico, Mario Rossi è una riga della tabella, da un punto di vista matematico (algebrico) Mario Rossi è un PUNTO in uno spazio bi-dimensionale. Tale astrazione permette anche di gestire immagini e testi (non-strutturati).

	Età	Altezza
Mario Rossi	35	175

A geometrical view of a p-dimensional dataset.

The space dimensions are here our dataset's columns, the vector's components are our dataset's row values.

P=6



Disclaimer: all the projections should be orthogonal to the axes, but a 6-dim space is NOT so easy to display....

Legenda:

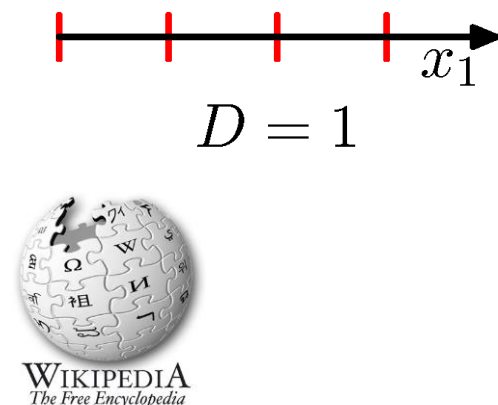


A vector in the space R^6 , that is a statistical unit

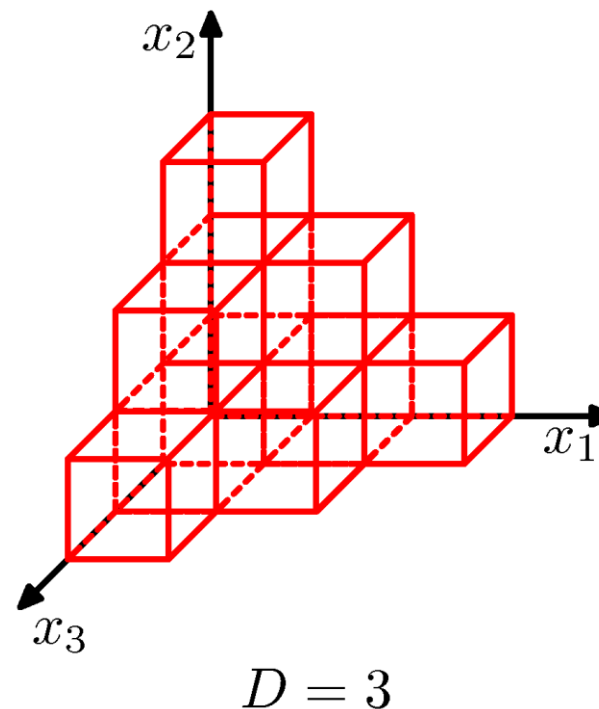
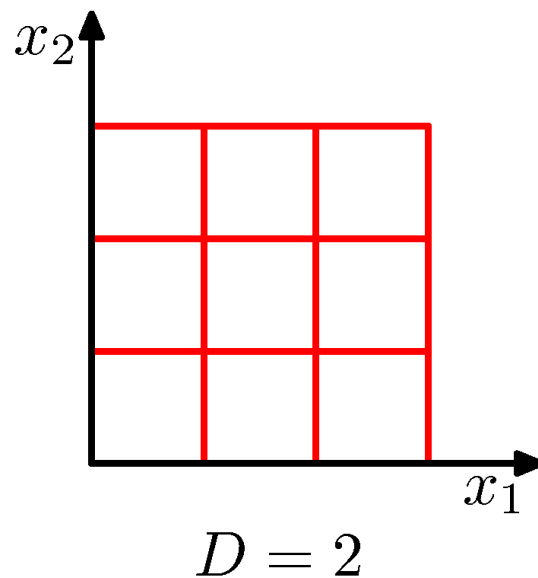


The vector's component along each dimension.

Al crescere del numero di dimensioni, lo spazio si fraziona sempre di più.



The curse of multi-dimensionality



Curse of dimensionality

From Wikipedia, the free encyclopedia

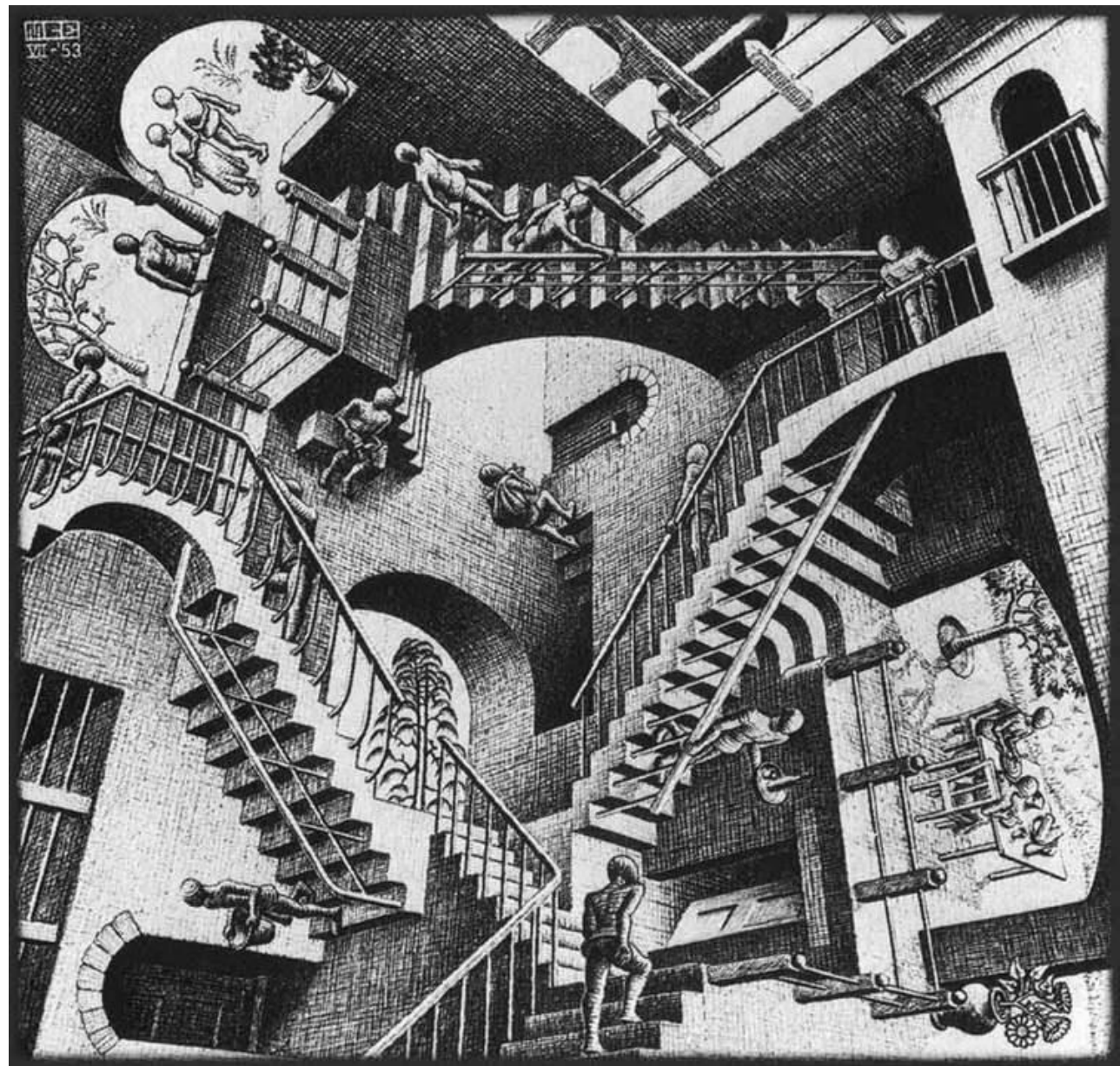
The **curse of dimensionality** refers to various phenomena that arise when analyzing and organizing data in [high-dimensional spaces](#) that do not occur in low-dimensional settings such as the [three-dimensional physical space](#) of everyday experience. The expression was coined by [Richard E. Bellman](#) when considering problems in [dynamic programming](#).^{[1][2]}

Dimensionally cursed phenomena occur in domains such as [numerical analysis](#), [sampling](#), [combinatorics](#), [machine learning](#), [data mining](#) and [databases](#). The common theme of these problems is that when the dimensionality increases, the [volume](#) of the space increases so fast that the available data become sparse. This sparsity is problematic for any method that requires [statistical significance](#). In order to obtain a statistically sound and reliable result, the amount of data needed to support the result often grows exponentially with the dimensionality. Also, organizing and searching data often relies on detecting areas where objects form groups with similar properties; in high dimensional data, however, all objects appear to be sparse and dissimilar in many ways, which prevents common data organization strategies from being efficient.

Uno spazio p-dimensionale

(oppure spazio lineare o spazio vettoriale di dimensione p). Nel ML questo spazio ha dimensione $k \leq p$ (con k = numero di colonne linearmente indipendenti della matrice di osservazioni $n \times p$)

esplorazione multi-dimensionale di articoli scientifici

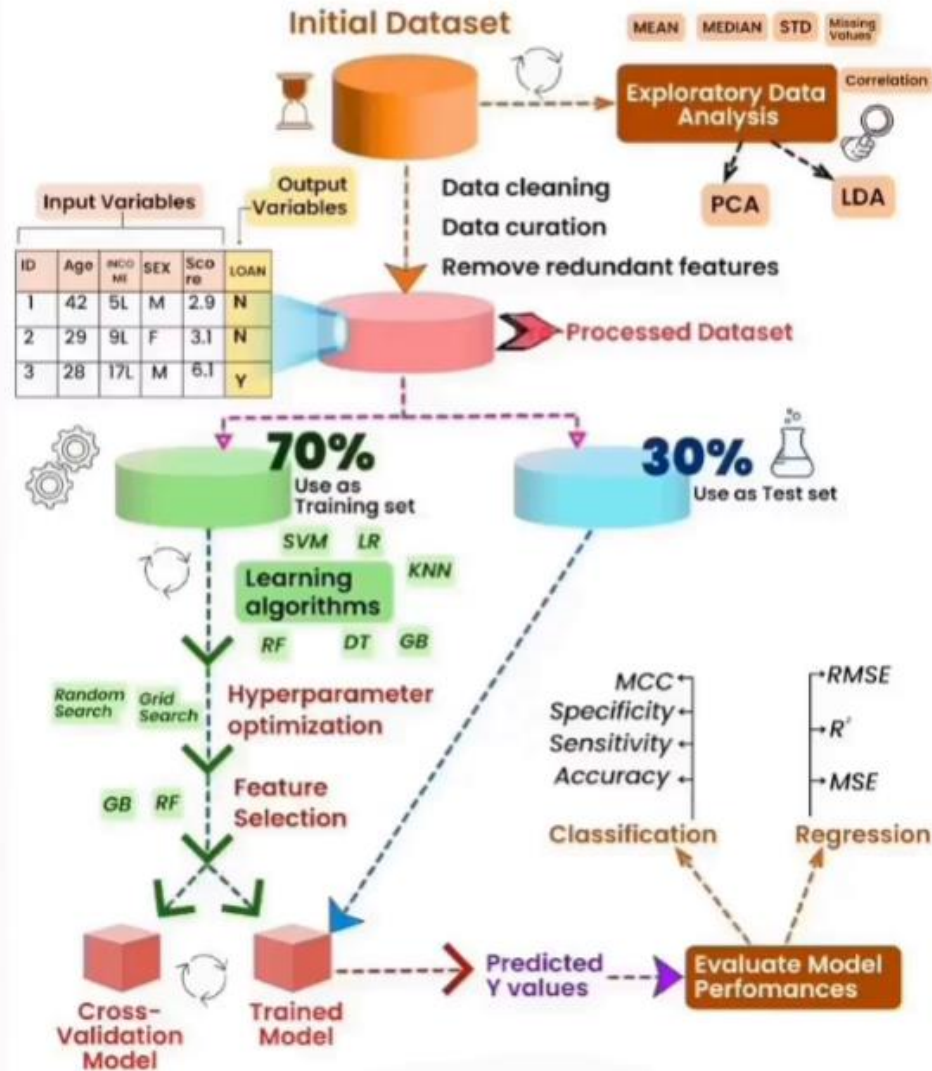


Famoso video sulla visualizzazione degli spazi ad alta dimensione e sul clustering. 3 esempi: persone, cluster parole e cluster cifre scritte a mano.

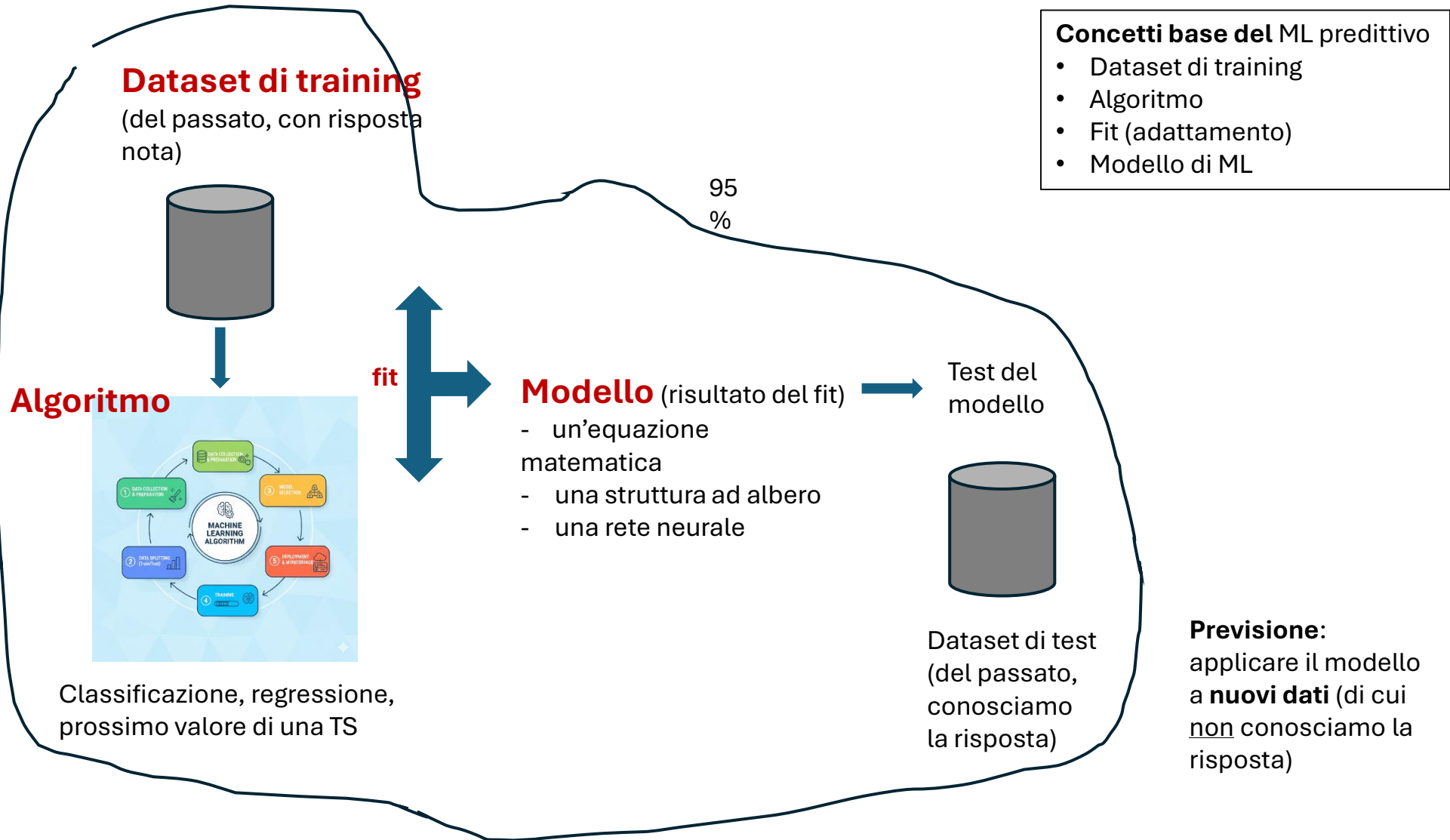
Il processo di ML



Building the Machine Learning Model

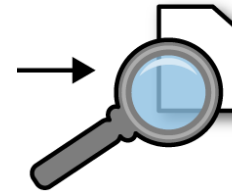


Il processo d ML



Dati storici

Job	Durat	Chec	Credit amoun	Purpos	Default
2	6	little	1169	radio/TV	no
2	48	mode	5971	radio/TV	no
1	12	little	2096	educati	yes
...	...	...	...	...	...

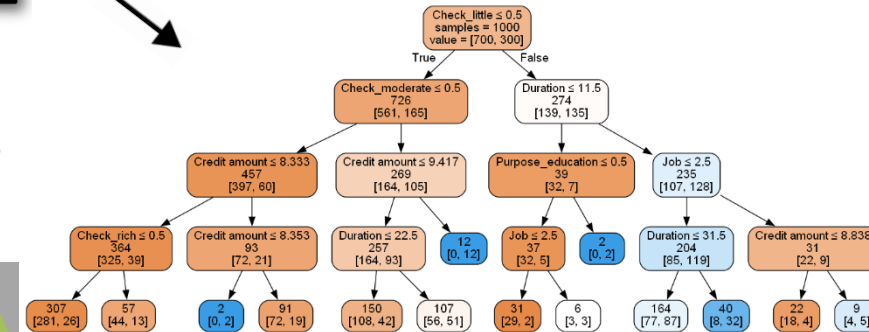


1 training del modello



2 test del modello

un albero,
un'equazione, una rete
neurale



il modello testato è messo a disposizione degli utenti

Data Scientis

Machine Learning

Utente

Utilizzo del modello

una nuova richiesta di prestito

Job	Durat	Chec	Credit amoun	Purpose	Default
1	10	little	2000	car	??

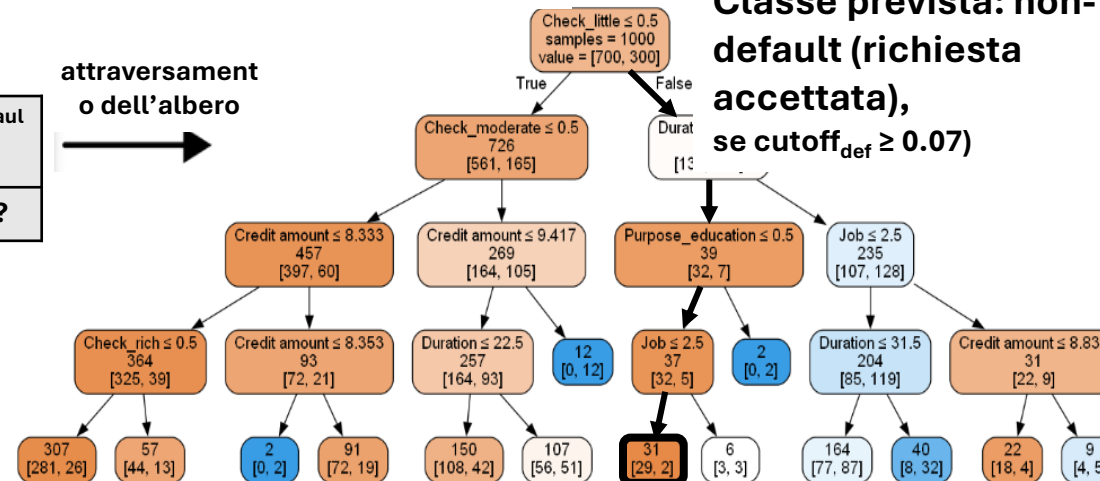
la nuova richiesta di prestito ha classe sconosciuta (l'affidatario restituirà il prestito alla sua scadenza?)

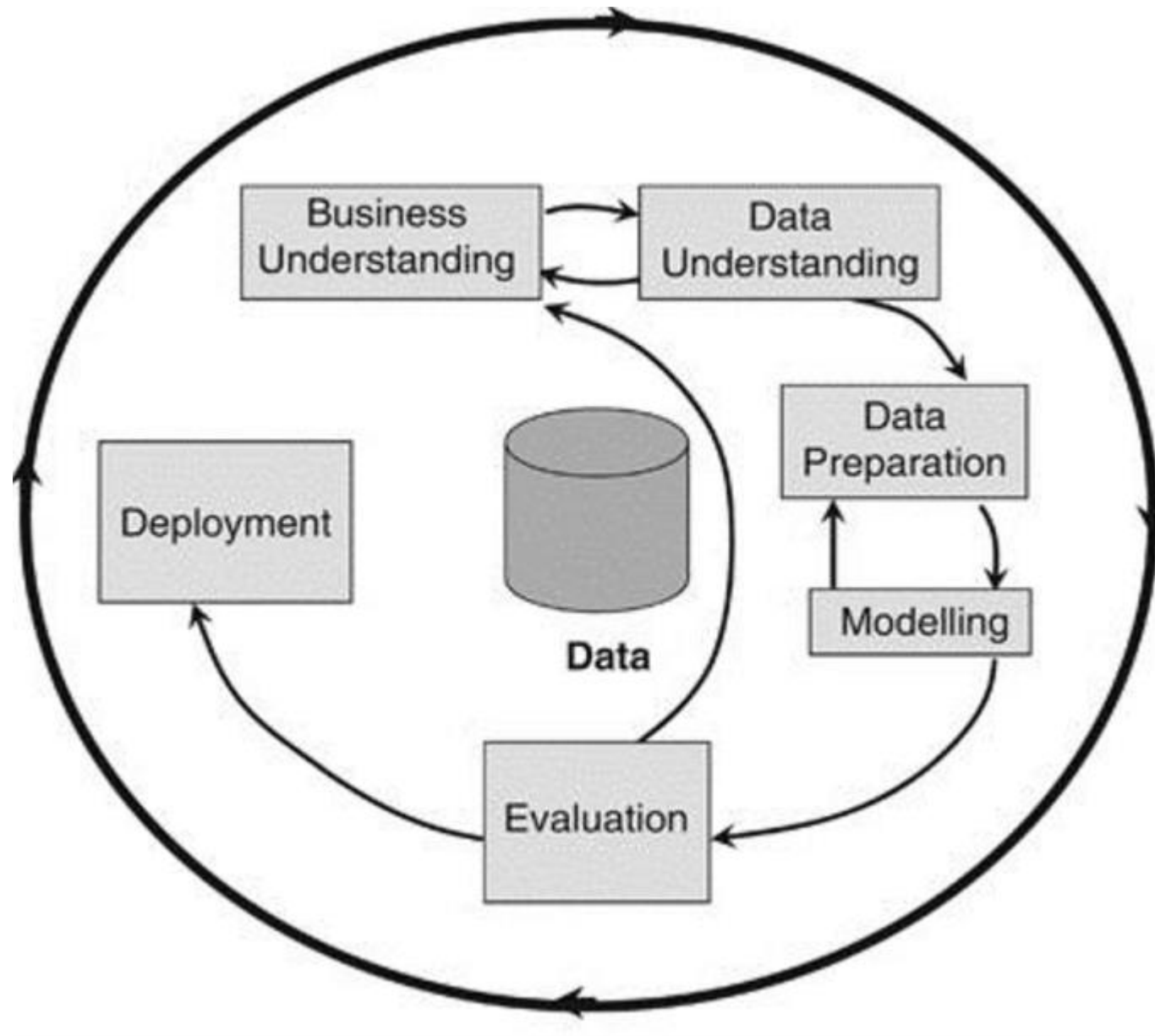
attraversament
o dell'albero

Il modello

Probabilità_{nondef}: 0.935 (29/31).

Classe prevista: non-default (richiesta accettata), se cutoff_{def} ≥ 0.07)



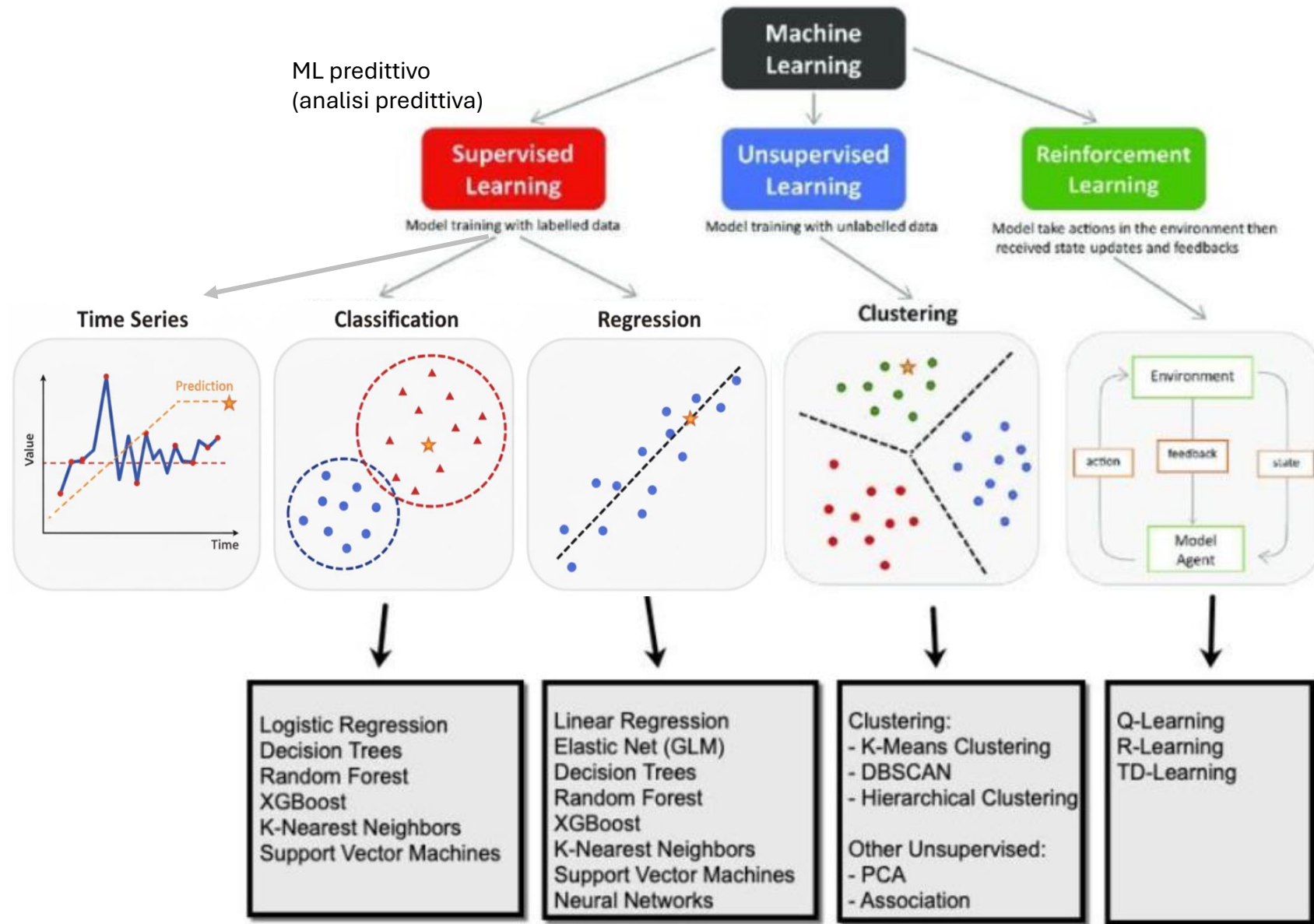


Crisp-DM Lifecycle

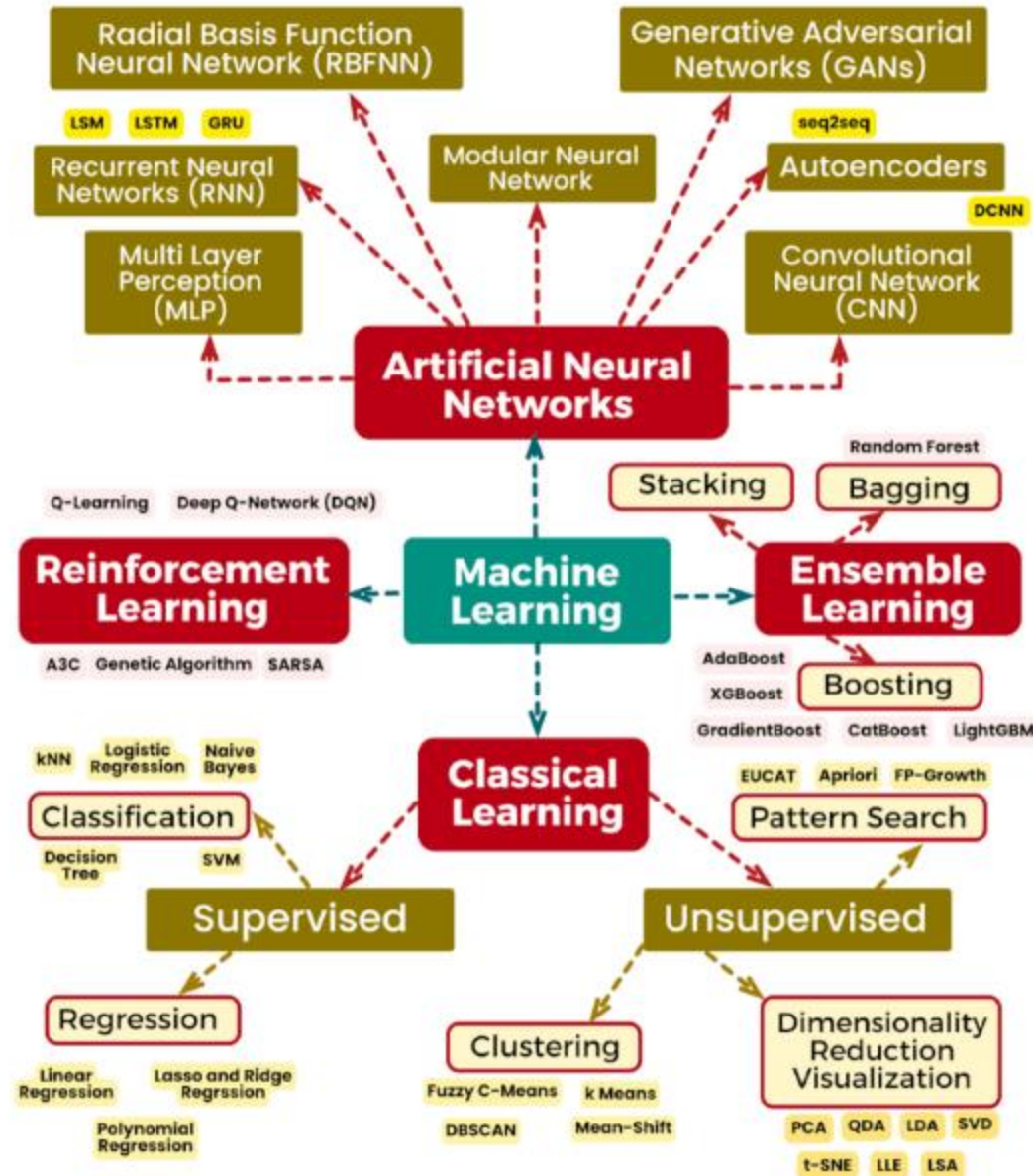
Apprendimento e algoritmi

3 Types of Machine Learning

(Every Data Scientist Should Know)



Machine Learning Algorithms

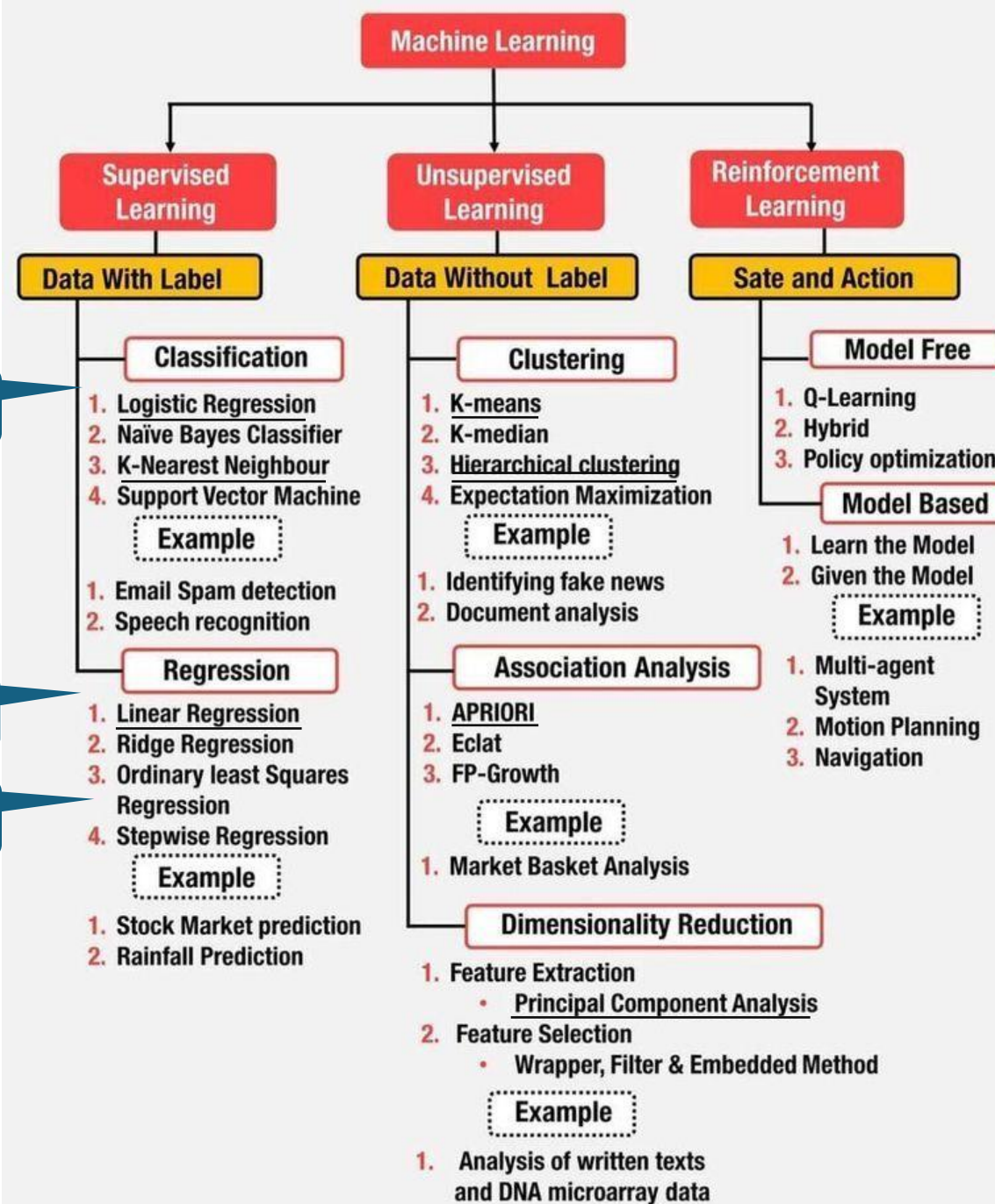


Per dati
strutturati

0. Alberi e
Fandom Forest

0. Alberi e
Fandom Forest

KNN (1/3 delle
volte è il migliore)



10
algoritmi
più diffusi

TOP MACHINE LEARNING MODELS



Support Vector Machines

Support vector machines (SVMs) are a type of machine learning algorithm that is used for classification and regression tasks, and finds the hyperplane that maximally separates the classes in the data.



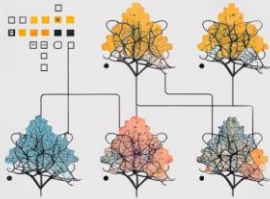
Neural Network

A neural network is a type of machine learning algorithm that is inspired by the structure and function of the human brain, and consists of interconnected processing nodes that are organized into layers.



Decision Tree

A decision tree is a type of machine learning algorithm that uses a tree-like model to make predictions based on the relationships between the features in a dataset. It works by dividing the data into smaller and smaller groups based on the values of the features, until it reaches a decision about the target variable for each group.



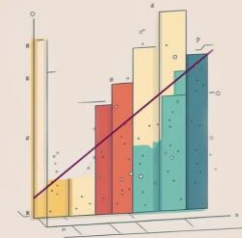
Random Forest

A random forest is a type of ensemble learning algorithm that trains multiple decision trees on random subsets of the data and then combines their predictions to make a final prediction. This can improve the performance of the model compared to using a single decision tree.



Nearest Neighbour

Nearest neighbor is a type of machine learning algorithm that makes predictions for a sample by finding the most similar samples in the training data and using their labels to make a prediction.



Linear Regression

Linear regression is a type of machine learning algorithm that is used to model the linear relationship between a dependent variable and one or more independent variables. It works by fitting a straight line (or hyperplane in higher dimensions) to the data that minimizes the distance between the points and the line. This line can then be used to make predictions on new data.

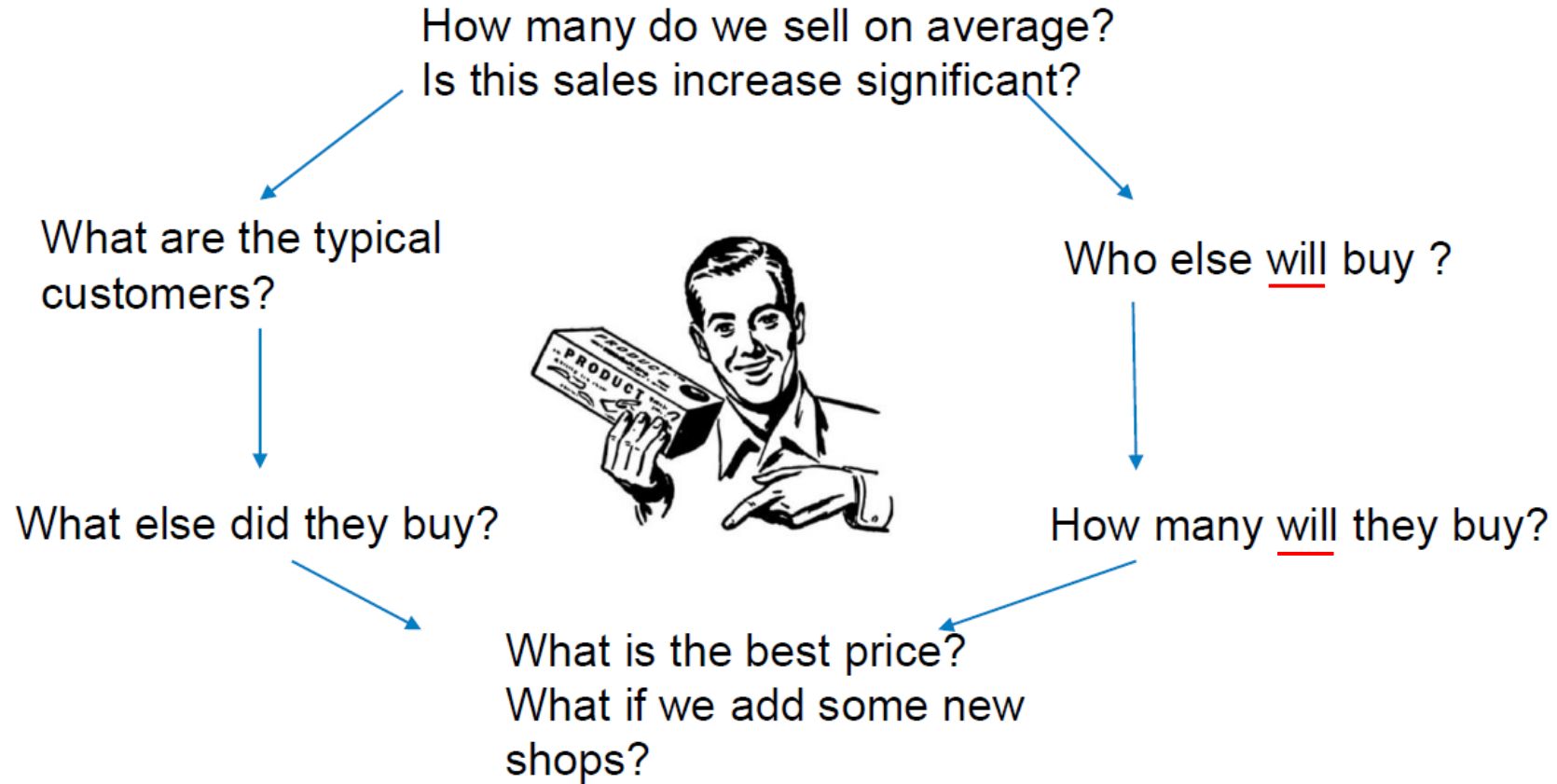


Logistic Regression

Logistic regression is a type of machine learning algorithm that is used for classification tasks, and models the probability that a sample belongs to a certain class using a logistic function.

Il problema «business»

Example: How to sell more of your favorite product?

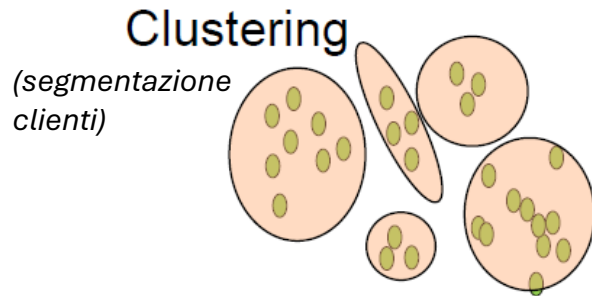


La tecnica ML di soluzione

(1:1 posizionale con la slide precedente)

$$Y = f(\text{reddito}, \text{zona}) \\ = k + w_1 x_{\text{reddito}} + w_2 x_{\text{zona}} + \text{reddito}^2$$

Example: How to sell more of your favorite product?

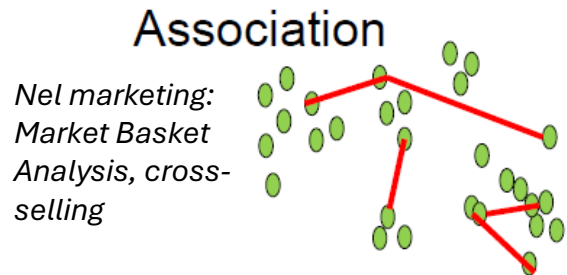
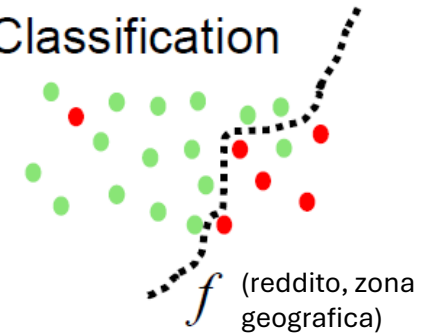


Reporting & Statistics

(nuovi termini: ML descrittivo, EDA)



Classification

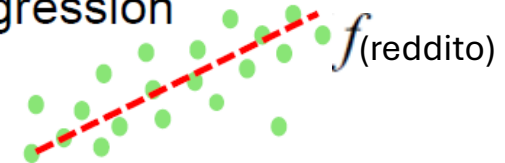


Operations Research

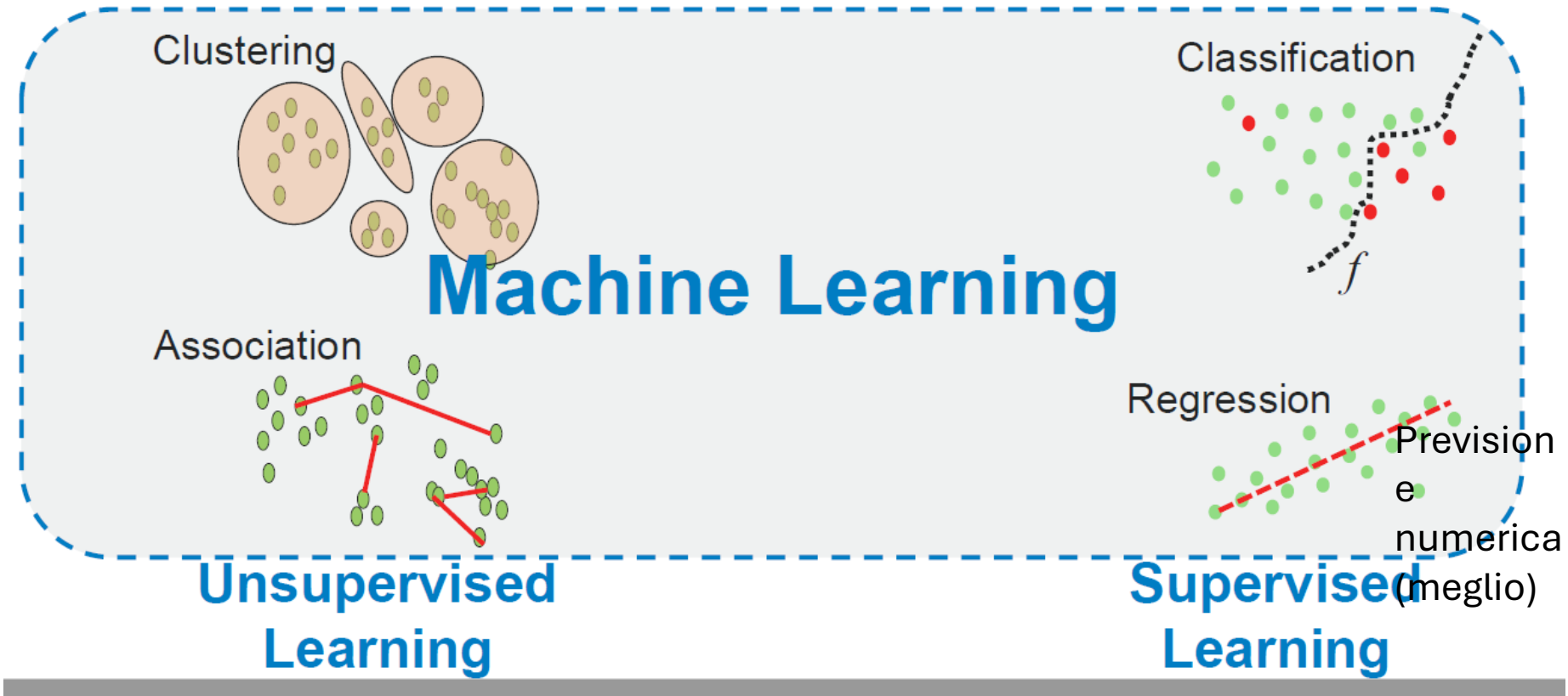


Analisi what-if

Regression



Example: How to sell more of your favorite product?



Supervised vs Unsupervised

Supervised learning: a person has to define the correct output for some portion of the data. Data is divided into training sets used for model building and test sets for validating the results.

What constitutes a good vs. bad credit risk?



"Apparently, you're well liked by your subordinates. That troubles me."

Unsupervised learning: the algorithm functions without an agent specifying the action to take on a given set of input. The algorithm derives the pattern.

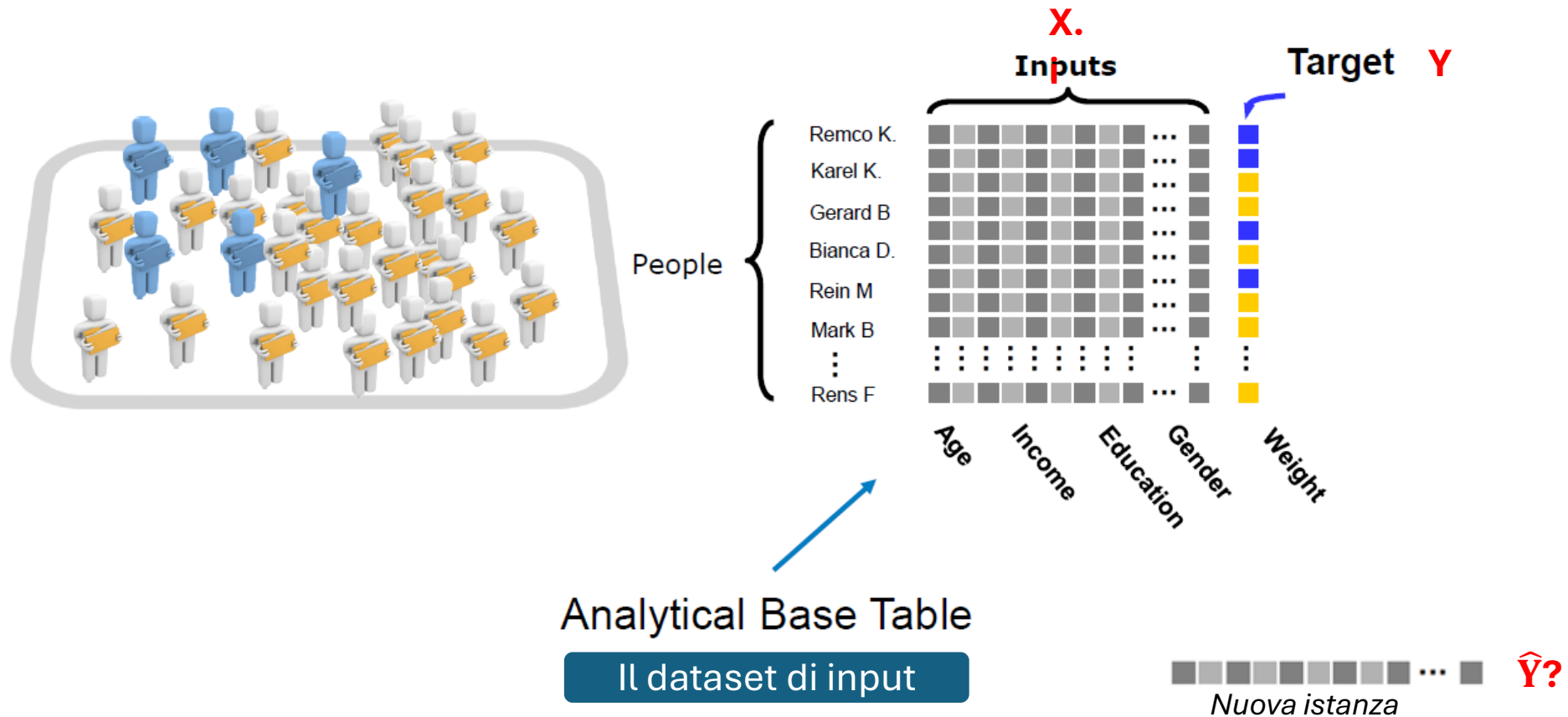
What is the best way to segment customers for marketing?

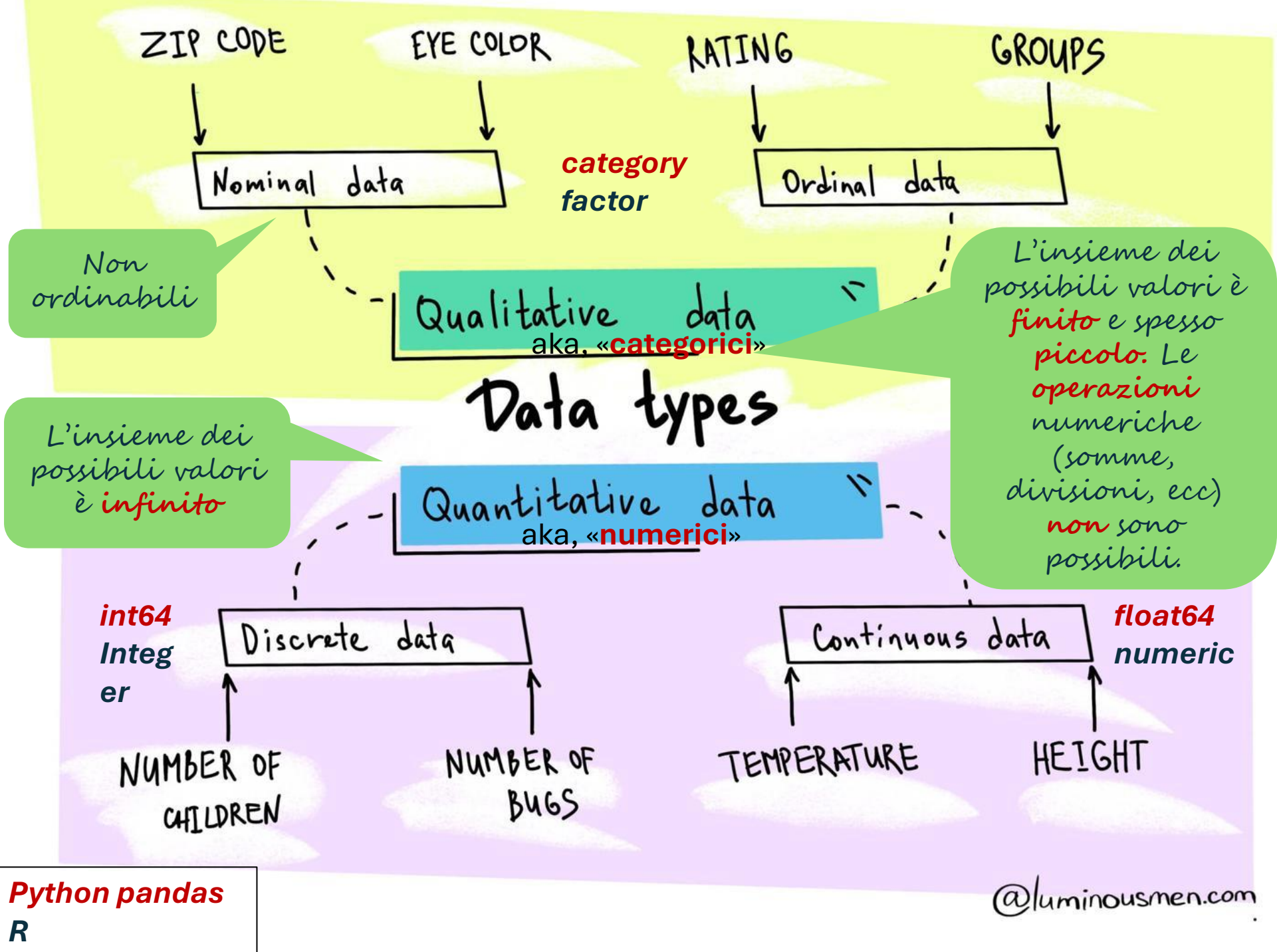


Predictive (Supervised) Modeling

How does this REALLY work....?

Discover patterns & relationships in data to predict an outcome/response

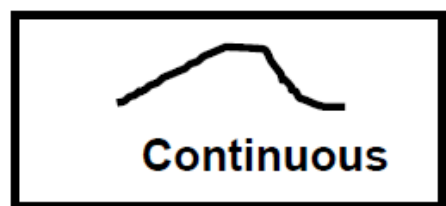




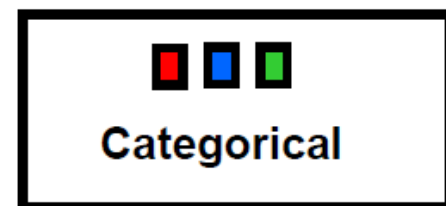
Target type & Modeling Techniques

Target/Response

Model type



Linear Regression
Regression Trees



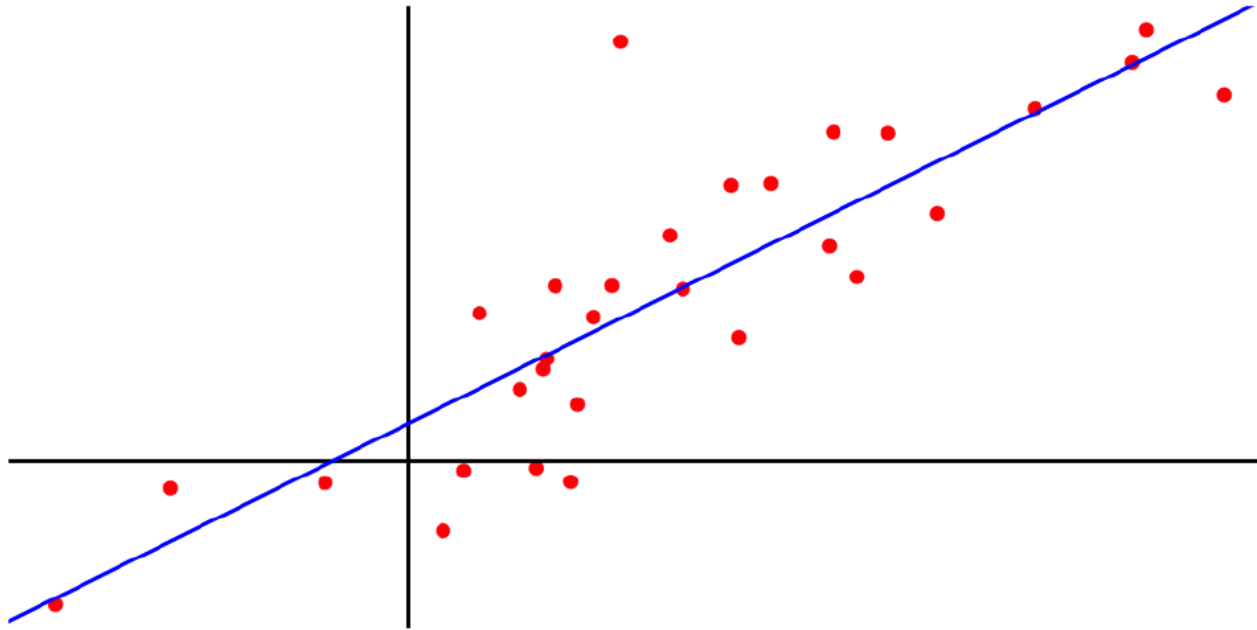
Logistic Regression
Decision Trees



Generalized Linear
Models

What's a Model?

Representation/abstraction of reality (“the map is not the territory”)



This is a model:

$$Y=a+bX$$

Supervised Machine Learning

(Classificazione)



Round things
look like this ...



Pointy things
look like this ...



More likely to be
pointy than **round**

Source: Gartner
ID: 465941 C

Unsupervised Machine Learning

(Clustering)



Blue Things



Yellow Things



Gray Things



Insight: Blue things tend to be pointy ...

Source: Gartner
ID: 465941_C

Questa struttura «latente» (il colore differente) è visivamente molto chiara nelle immagini; è un esempio ideale, è facile e comprensibile clusterizzare per colore; in genere le strutture latenti non sono così chiare nei dati

Il punto didattico della slide è:

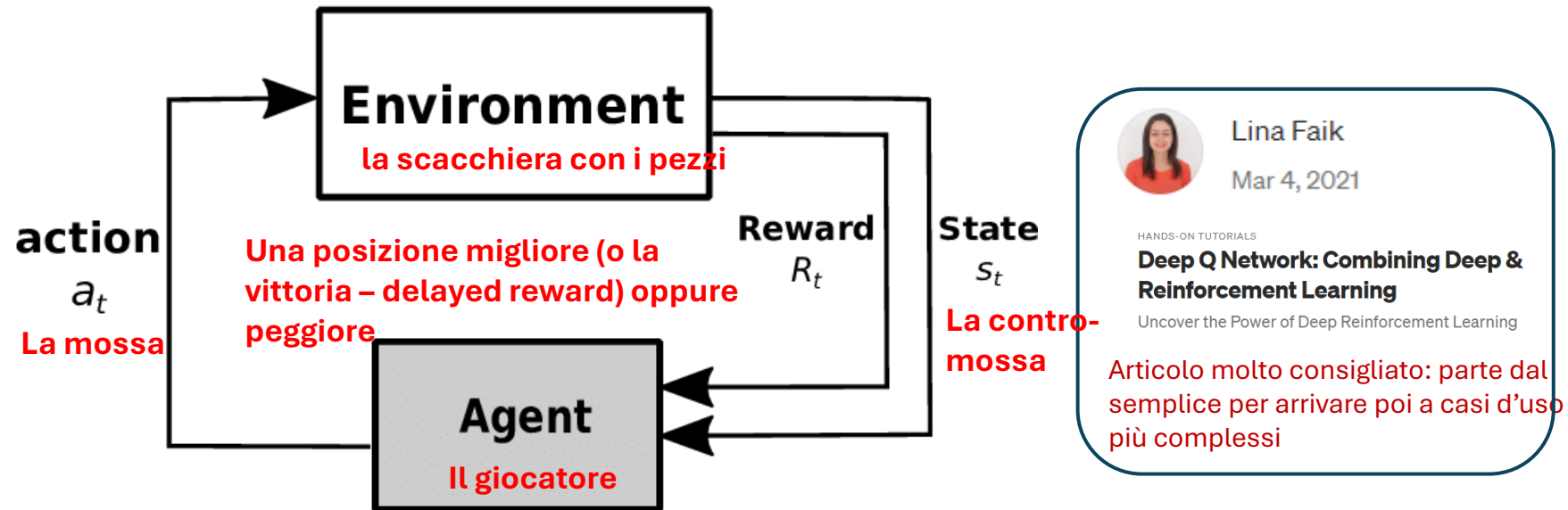
sopra: *ho etichette* → *imparo a distinguere tra round e pointy*

sotto: *non ho etichette* → *scopro che esistono 3 famiglie*

Semisupervised learning (ISLR, p. 28):

- A volte i predittori/feature del training set possono essere osservati più facilmente (cioè con «costo» minore) della label, che spesso deve essere attribuita manualmente;
- Si supponga di conoscere (anche) la risposta (la label) per $n < k$ osservazioni, e per le restanti $(n - k)$ di conoscere solo il valore dei predittori/feature;
- Il confine tra supervisionato e non-supervisionato è qui meno chiaro;
- Serve allora un classificatore (in generale un metodo di ML) che sappia incorporare entrambi i tipi di osservazioni nello stesso modello.

Reinforcement Learning (es. il giocatore di scacchi)



Altro esempio intuitivo: l'addestramento di cani o cavalli (gli agenti), ai quali sono forniti ordini e eventuali ricompense (zuccherini) per incentivare i giusti comportamenti.



[Link al video YouTube](#)

[Link alla clip](#)



The Blondoner

@TheBlondoner · 2670 iscritti · 91 video

Un punto di vista diverso sul mondo ● ...altro

[instagram.com/iamtheblondoner](https://www.instagram.com/iamtheblondoner)

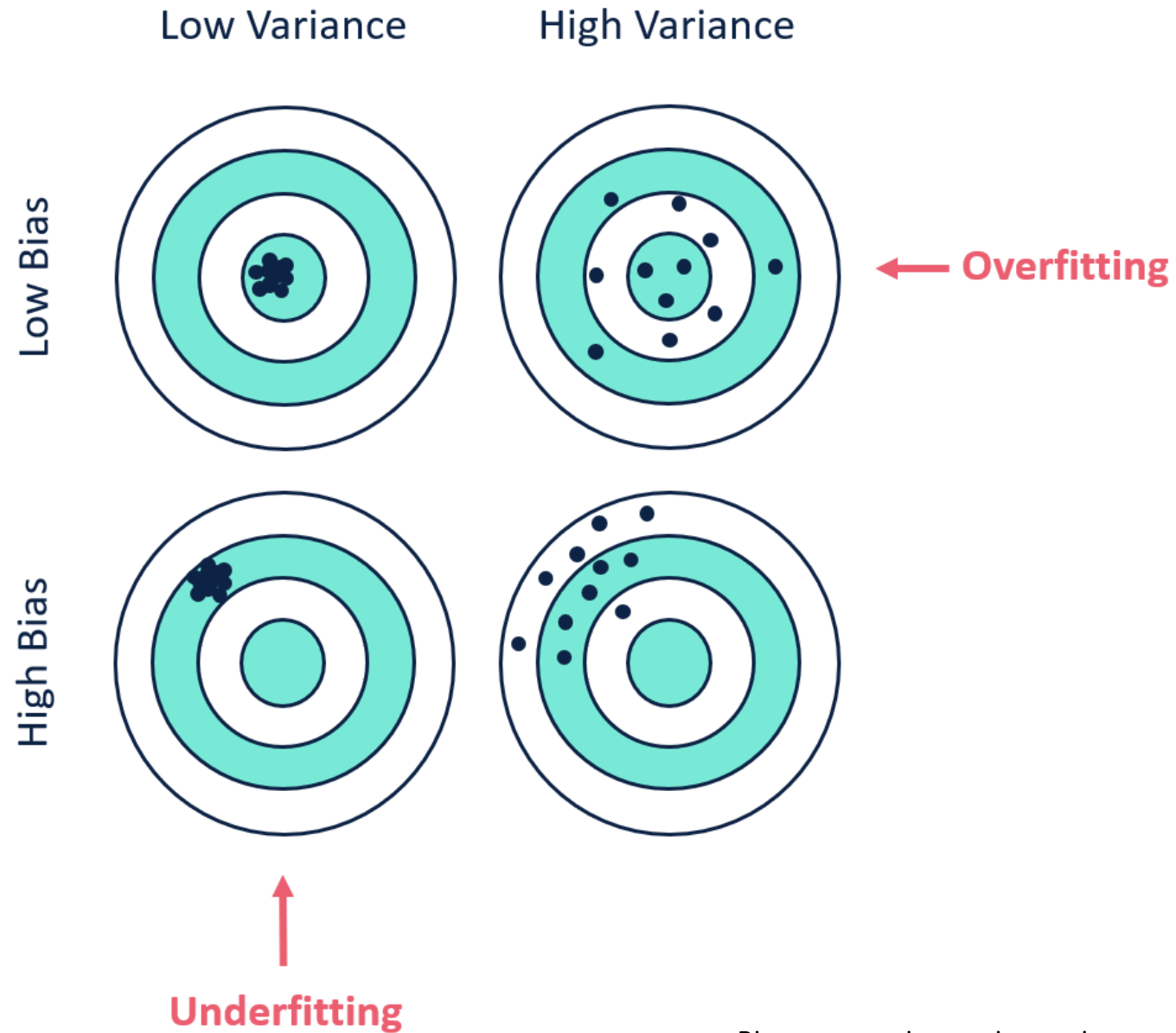
Iscriviti

Abbonati

Bias e Varianza

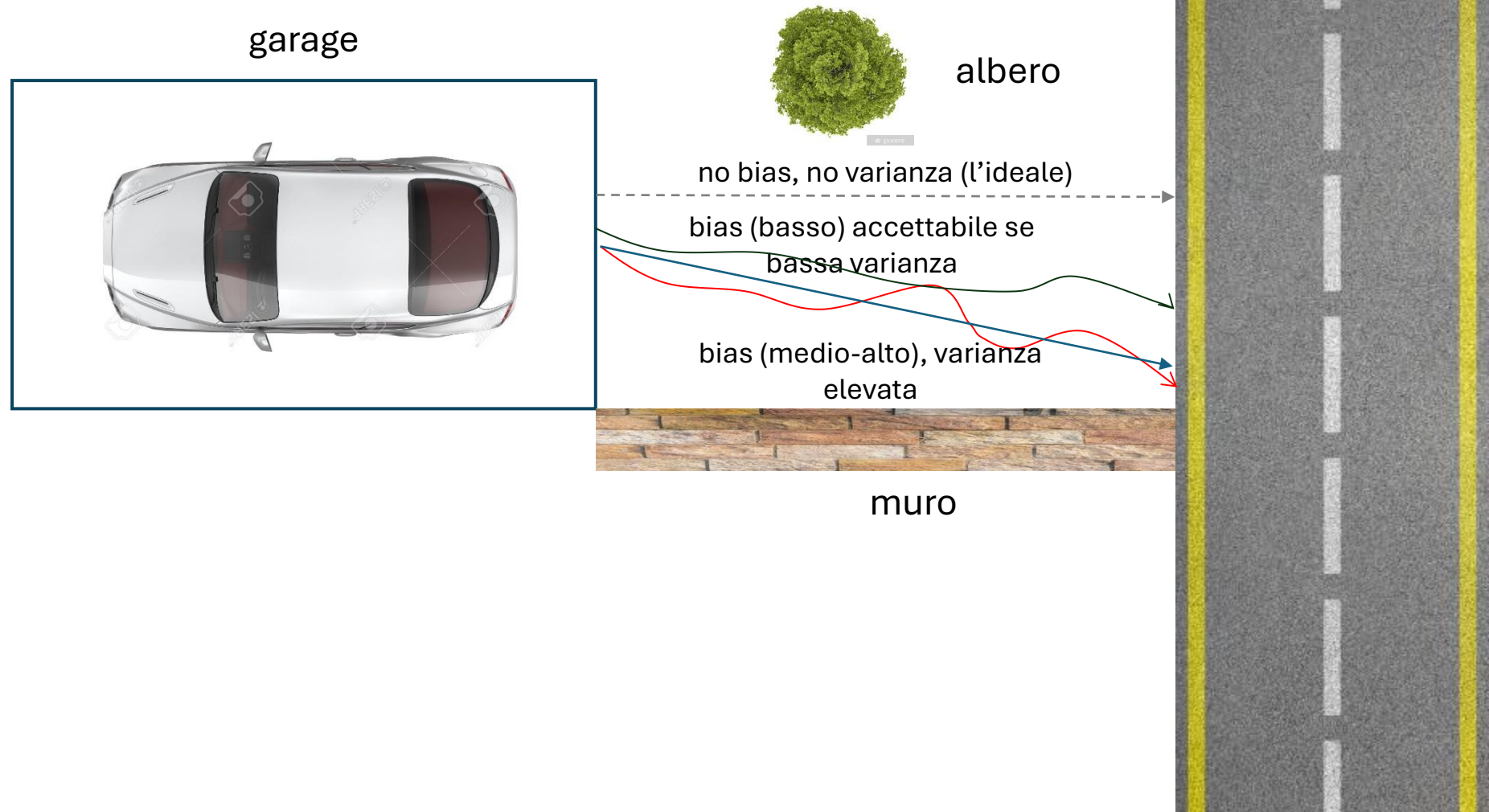
Overfitting e Underfitting

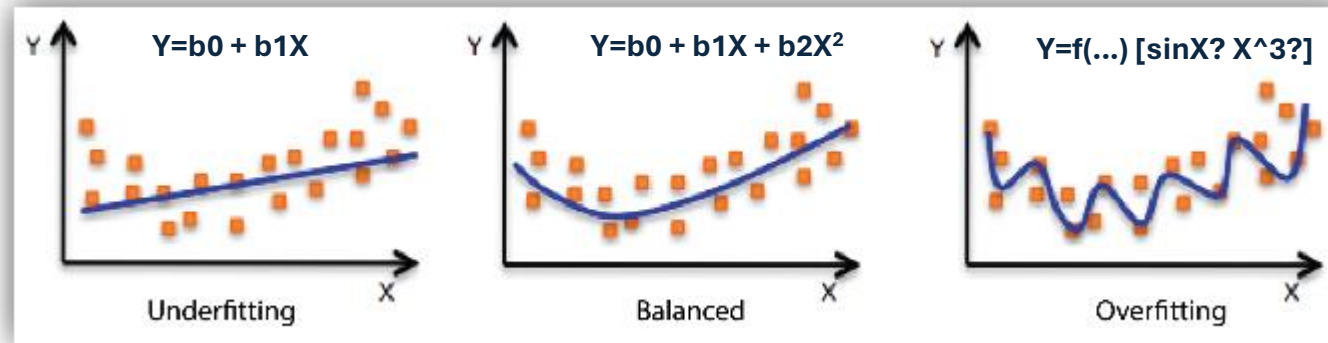
Bias vs. Variance (in generale, a prescindere dal ML)



Bias = errore sistematico; varianza = variabilità

Bias vs. Variance (in generale, a prescindere dal ML)





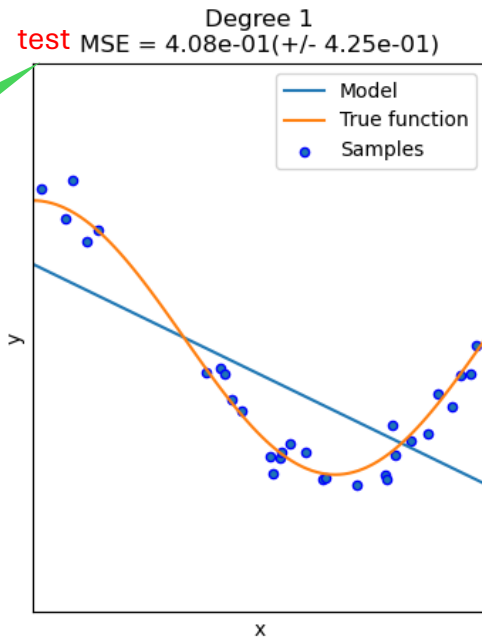
Fonte: scikit-learn.org

Bias abbastanza basso, varianza abbastanza bassa

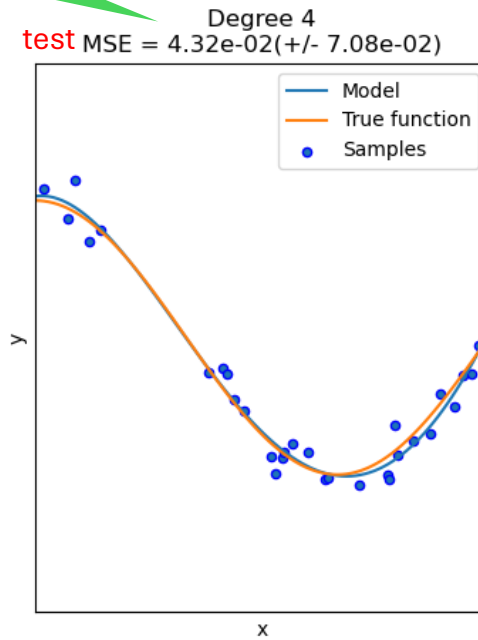
Bias molto basso, varianza altissima

Fonte: Amazon Machine Learning

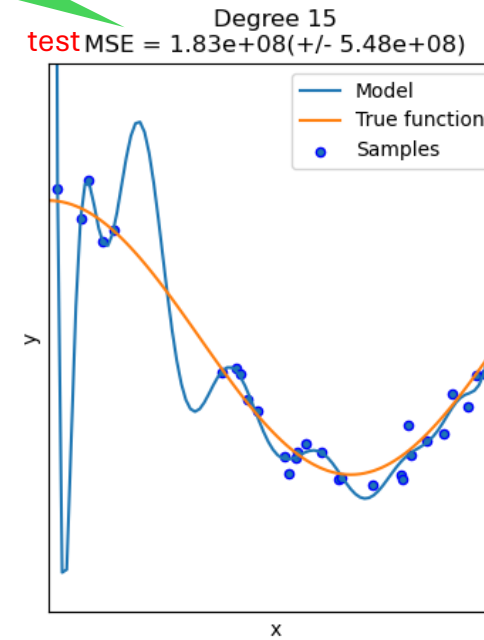
Bias alto, varianza bassa



Funzione vera: $Y = \beta_0 + \beta_1X + \beta_2X^2 + \beta_3X^3 + \text{eps}$
Modello stimato: $Y = b_0 + b_1X$

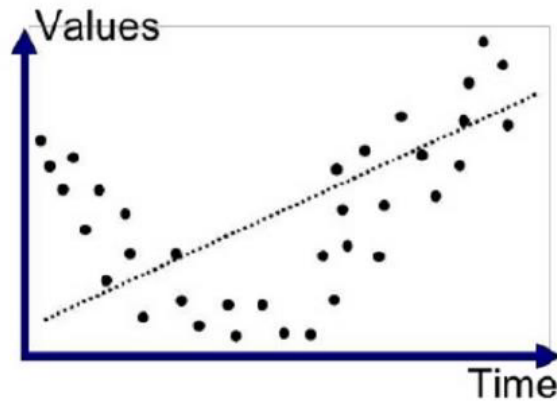


Funzione vera: $Y = \beta_0 + \beta_1X + \beta_2X^2 + \beta_3X^3 + \text{eps}$
Modello stimato: $Y = b_0 + b_1X + b_2X^2 + b_3X^3$

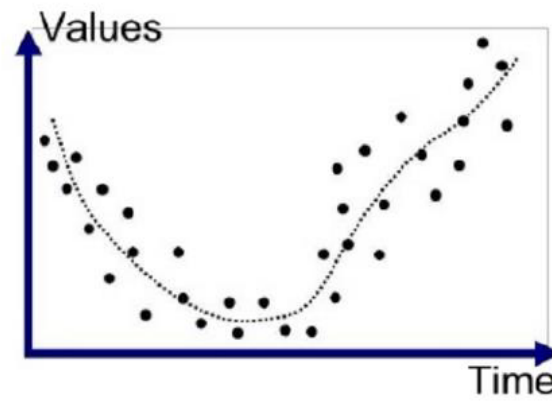


Funzione vera: $Y = \beta_0 + \beta_1X + \beta_2X^2 + \beta_3X^3 + \text{eps}$
Modello stimato: $Y = b_0 + b_1X + b_{15}X^{15}$

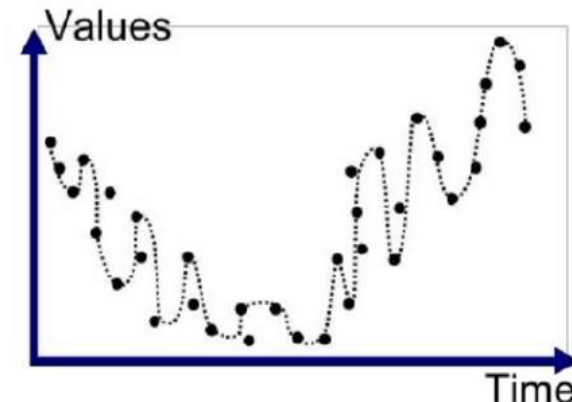
Right Fit



high bias
low variance



medium bias
medium variance



low bias
high variance

Underfit Overfit

Pattern non
«generalizzabile» ad
altre istanze di dati
(Witten, DSfB)



Source: Joannes Vermorel, 2009-04-19, www.lokad.com

Trade-off bias vs varianza ineliminabile, a meno di
casi particolari (gestiti con alcune reti neurali)

Fonte: Jos van Dongen

Beware of Overfit!

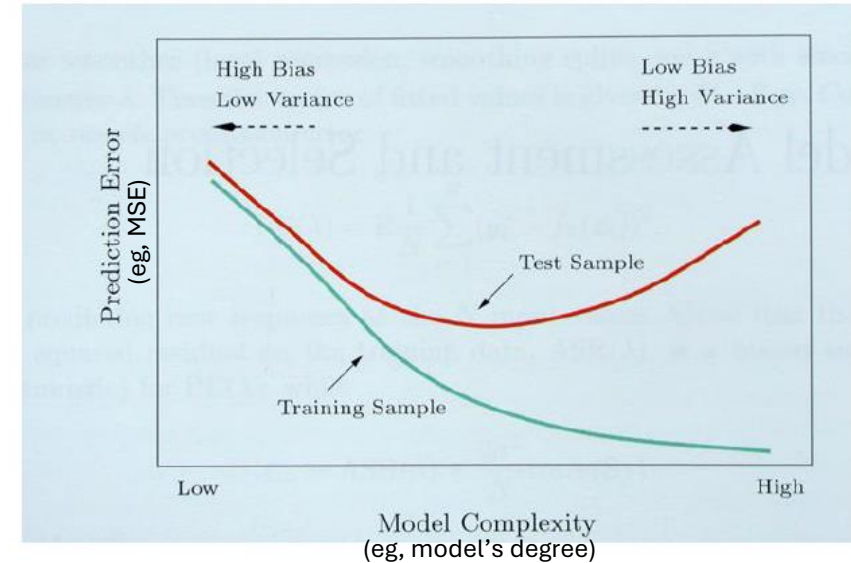
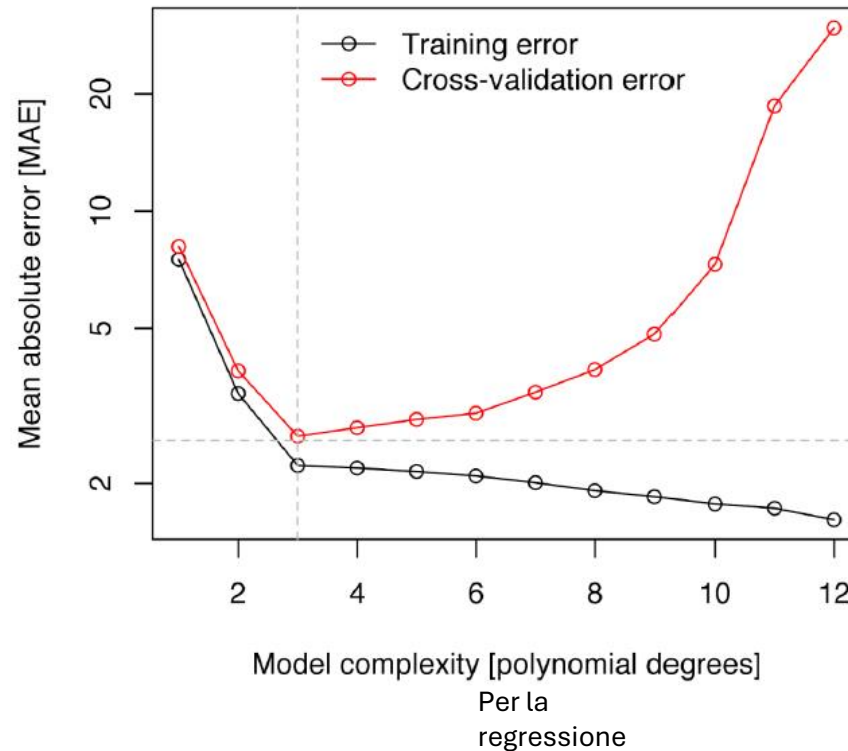
Da cosa è data la complessità di un modello?

Regressione: il numero e grado dei predittori.

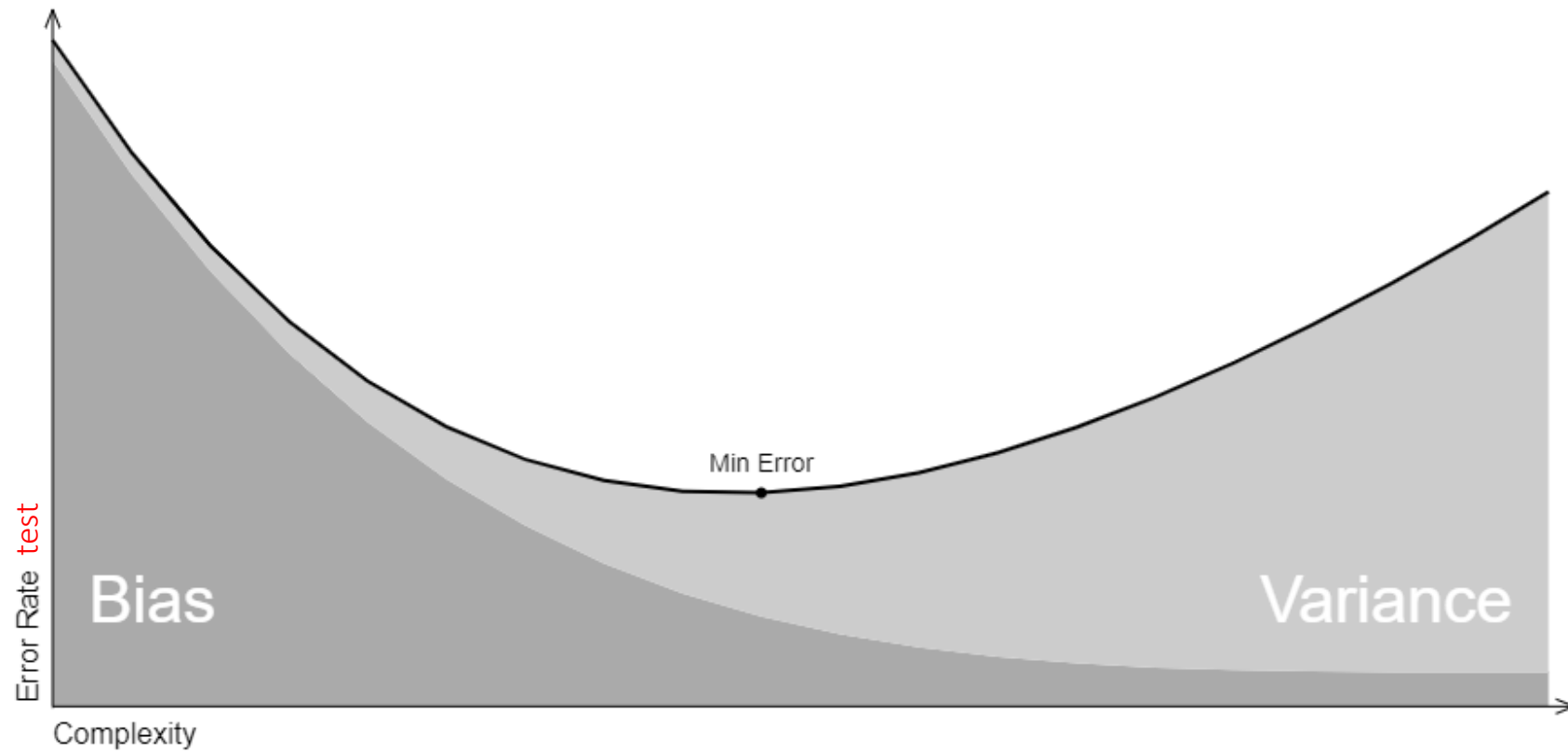
Alberi: profondità, num. min istanze per foglia, num. min. istanze nodo per suddividerlo

KNN: $1/K$

Reti neurali: #layer nascosti, # neuroni, funzione di attivazione, ecc



Source: Hastie, Tibshirani, Friedman "Elements of Statistical Learning" 2001



minimum node size
= 89



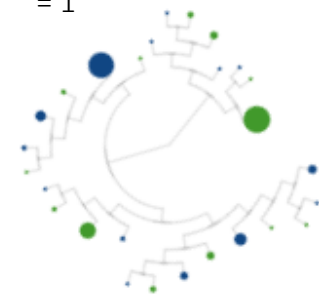
minimum node size
= 62



minimum node size
= 5/6 (sweet spot)



minimum node size
= 1



Questa è la regola generale del Machine Learning. In base a studi recentissimi sembra che alcuni tipi di reti neurali riescano ad abbassare il bias senza aumentare la varianza?! – Da verificare